Project	**FEED for Gas Gathering Pipelines & Associated Infrastructure**
Client	**Brass Fertilizer and Petrochemical Company Limited**
Originating Company	**Epic Atlantic Limited**
Document Title	**FEED Package Overview**
Document Number	**EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-001**
Document Revision	R01
Document Status	Issued For Review
Coordinator	Aloy Duru
Security Classification	Restricted
Issue Date	**10 Aug 2022**

FEED Package Overview

R01	10 Aug 2022	Issued for Client Review	W. Irem	M. Ekpunobi	A. Duru	
Rev #	**Issue Date**	**Status Description**	**Originator**	**Reviewer**	**Approver (Epic)**	**Approver (BFPCL)**

Revision History

Rev. No.	Date	Description of Revision
P0	10 Aug 2022	For Review

Revision Philosophy:

a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
e. All approved “As-Built” documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the “As-Built” document to Z02).

Summary Page

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	**Project No.** 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	**Doc No.** EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-001-R01

The administration of this document is in respect of FEED Package Overview for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The document, in consideration of relational design elements & concepts, is a summary of the of the key features of the FEED, and its specific deliverables required by regulators, - which collectively forms the technical and statutory basis for the project's Financial Close, and the subsequent EPC processes implementation.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

Related to this scenario of two phases, is the opportunity to flow 1 BSCFD through the pipeline system to the 1 BSCFD gas hub proposed at Akabel as a separate project.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the key components of the key components of the FEED.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES	
Document No.	Description
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-001-R01	Basis of Design
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-0011-R01	Design and Operating Philosophy
EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
	Materials Specification & Corrosion Management
	Flow Assurance Report
	FISAS ESIA Draft Report (November 2020)

Table of Contents

Project Information, Terms, Abbreviations, and Definitions

Project Information – General

1.0	Employer	BRASS FERTILIZER & PETROCHEMICAL COMPANY LIMITED (BFPCL)
2.0	PMC	TATA CONSULTING ENGINEERS LIMITED (TCE)
3.0	EPCC (CONTRACTOR)	CHINA TIANCHEN ENGINEERING CORPORATION (TCC)
4.0	Project Title	Gas Processing and Methanol Complex
5.0	Location	Akabel, Brass Island. The project site is about 85 km South West of Port Harcourt, 70 km South of Yenagoa and 90 km West of Bonny Island.
5.1	State	Bayelsa State, Nigeria
5.2	Nearest Air Port and distance	Port Harcourt Airport (PHC), 80 km Yenagoa Airport (BIA), 75 km
5.3	Nearest Port	Port Harcourt (Onne)
5.4	Nearest Railway Station & distance	-
5.5	Highway and distance	Connected through water way only. Road connection does not exist.
5.6	Important Town and distance	Yenagoa, Nigeria& 67 km
5.7	Altitude Above Mean Sea Level	9 m
5.8	**Atmospheric pressure**	
	Average	**1,013 mbar**
6.0	Ambient Air Temperatures	**(Refer Note-5,6)**
6.1	Maximum	38°C
6.2	Minimum	18°C
6.3	Average	23°C
6.4	Summer Design Wet Bulb Temp	28°C
6.5	Winter Design Wet Bulb Temp	15°C
6.6	MDMT	15°C

Project Information – General (Contd)

6.7	Swamp temperature	25°C (23°C to be used wherever a margin is required)
6.8	Ground temperature	25 - 27.5 °C
6.9	Black-bulb temperature:	80°C
7.0	Relative Humidity	
7.1	Maximum	95 %
7.2	Minimum	51 %
7.3	Average	73 %
8.0	Rainfall Data	
8.1	Average annual rainfall	3,800 mm
8.2	Max daily rainfall:	180 mm
8.3	Mean max. Hourly rainfall	100 mm
9.0	Transport	
9.1	Name of the highway near which the plant is located	Connected through water way only. Road connection does not exist.
9.2	Access road	-
9.3	Railway (Gauge)	-
10.0	Wind Data	
10.1	Design Wind Velocity	35.6 m/sec (Note-1)
10.2	Wind Direction	Note-2
11.0	Seismic Data	
11.1	Earthquake Zone	Note-3
12.0	High Flood Level (HFL)	*
13.0	Natural Ground Level	*
14.0	Slope	*
15.0	Drainage	*
16.0	Design temperature for Electrical equipment	40°C

Definition of Terms

TERMS	DEFINITIONS
BFPCL	Brass Fertilizer & Petrochemical Limited.
SPDC	Shell Petroleum Development Company Nig. Limited
Employer	Party (Usually BFPCL) that initiates the project and ultimately pays for its design and construction. The Employer will generally specify the technical requirements. The Employer may also include an agent or consultant authorized to act for and on behalf of, the Employer.
Engineer	The party that carries out the FEED Engineering of the project.
Manufacturer/Supplier	Party, which manufactures or supplies equipment and services to perform the duties specified by the Contractor.
Shall	The word 'shall' indicate a mandatory requirement.
Should	The word 'should' indicate a recommendation.
Measuring Range	Range defined by LRL and URL within which the limits of uncertainty of the measuring instrument are specified (the capability of the instrument).
On-Line Instruments	Instruments connected to process and utility lines or equipment via primary isolation valves.
Project	Front End Engineering Design for Gas Gathering Pipelines (Phase1)

Abbreviations

Abbreviation	Meaning
AASHTO	American Association of state Highway Transportation Officials
AGEF	Above Ground End Facilities
ALARP	As Low as Reasonably Practicable
API	American Petroleum Institute
ASTM	American Society of Testing and Materials
BOD	Basis Of Design
CP	Cathodic Protection
CPT	Cone Penetrometer Test
CTMS	Custody Transfer Metering Station
DB&B	Double Block and Bleed
DCQ	Daily Contractual Quantity

Abbreviations (contd.)

Abbreviation	Meaning
DEP	Design and Engineering Practice
ESD	Emergency Shutdown
ESDV	Emergency Shutdown Valve
FEED	Front End Engineering Design
FGS	Fire, Gas and Smoke detection systems
GPP	Gas Processing Plant
GTG	Gas Turbine Generators
HAZID	Hazard Identification
HAZOP	Hazard Operability
HEMP	Hazards and Effects Management Process
HFE	Human Factors Engineering
HPU	Hydraulic Power Unit
HRA	Health Risk Assessment
HSES	Health, Safety, Environment and Security
HSSE & SP	Health, Safety, Security, Environment and Social Performance
IEC	International Electro-technical Commission
ISPM	Industry Standard Process Modules
JV	Joint-Venture
LCD	Liquid Crystal Display
LOI	Local Operator Interface
LRL	Lower Range Limit
LRV	Lower Range Value
LIRA	Logistics and Infrastructure Resource Assessment
MAC	Main Automation Contractor
NAG	Non-Associated Gas
OPC	OLE for Process Control
OPPS	Overpressure Protection System

Abbreviations (contd.)

Abbreviation	Meaning
PCD	Process Control Domain
PDMS	Plant Design and Management System
PEPF	Package Engineering, Procurement, and Fabrication
PIG	Pipeline Inspection Gauge
PVC	Polyvinyl Chloride
RTR	Reliability Test Run
SCE	Safety critical elements
SI	International System Units
SIL	Safety Integrity Level
SRB	Sulphate Reducing Bacteria
SIS	Safety Instrumented System
SWB	Steel Wire Braiding
URV	Upper Range Value
USC	US Customary Units
XLPE	Cross Linked Polyethylene
TEG	Tri-ethylene Glycol
TQ	Top Quartile

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

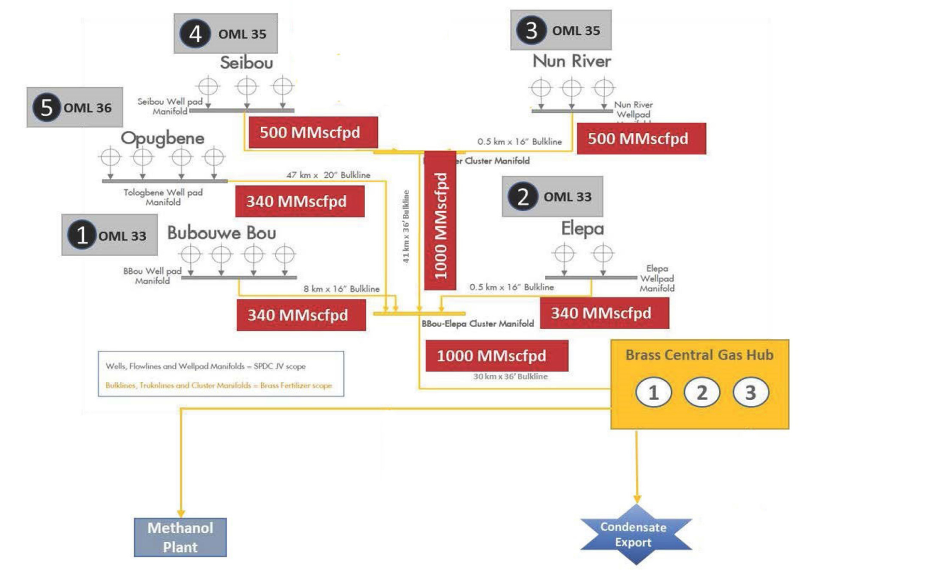
TCE is the Project Management Consultant for the project, and TCC the EPC Contractor.

1.2. The Project Phases

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later; and the related 1 BCFD gas hub opportunity much later.

<u>The Nodal Gas Gathering Schematics</u>



Gas Gathering from five SPDC Nodes

Figure 1-1: Fields & Nodes covered in Phase 1 & 2 Gas Gathering

The Nodal Gas Gathering Schematics (contd.)

Gas Gathering Concept, Phase-1 (BBOU & ELEP): E-MF at ELEP and S-MF at BBOU

Nodal Gas Gathering at BBOU & ELEP with BBOU Node Gas Joining ELEP Node for Onward Transportation to the Gas Processing Plant at AKAB

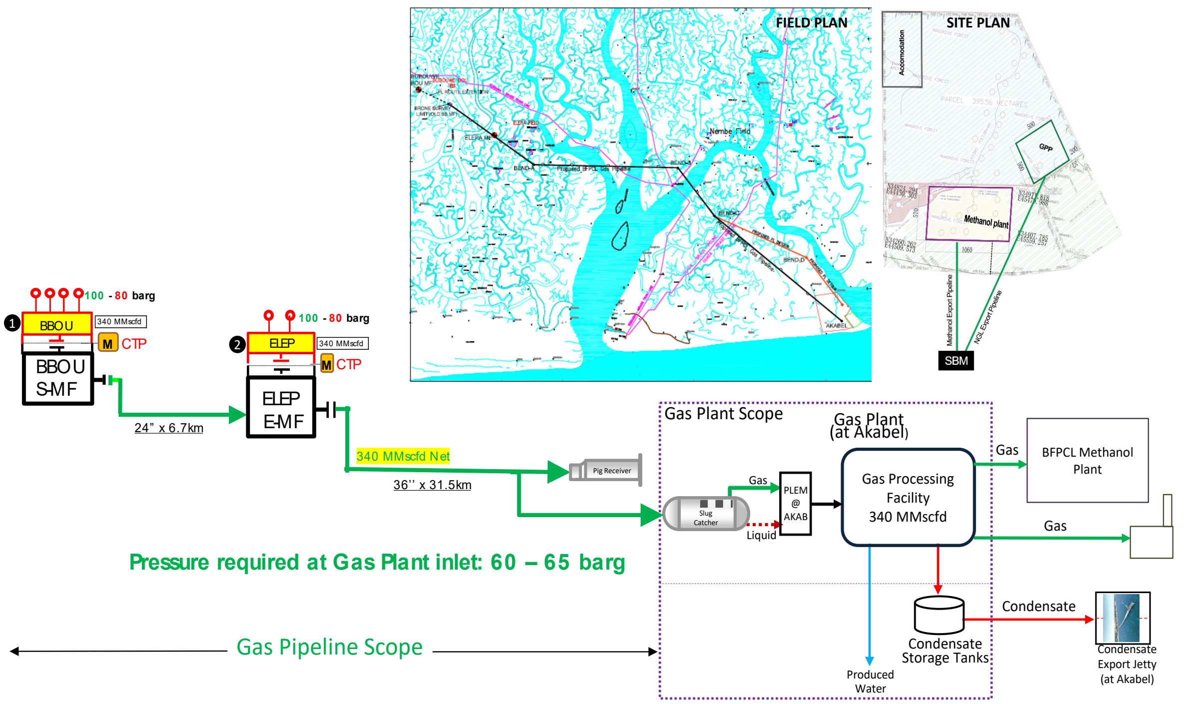


Figure 1-2: Gas Gathering Concept, Phase 1

Wholistic Phase 1 & 2 Gas Gathering Concept

Nodal Gas Gathering at BBOU, ELEP, NUNR, SEIB, & OPUG, with BBOU, NUNR & SEIB & their Upstream Nodes Joining ELEP Node for Onward Transportation to the GPP at AKAB

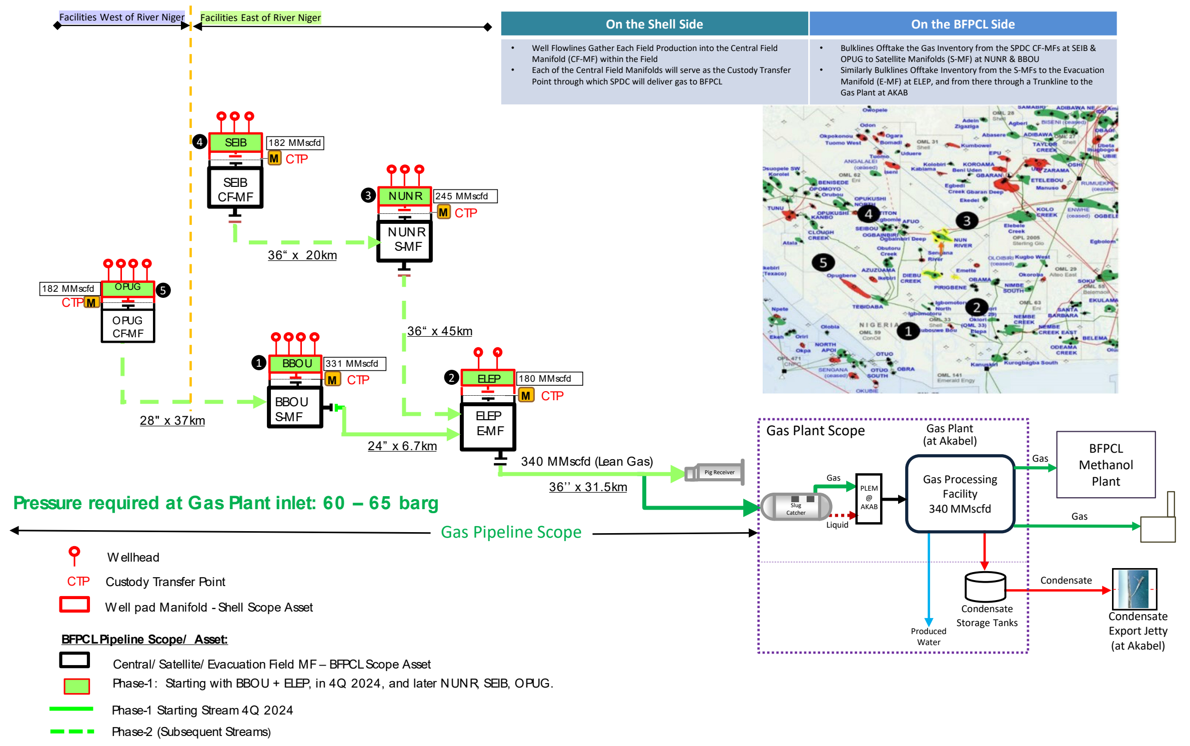


Figure 1-3: Wholistic Gas Gathering Concept, Phase 1 & 2

1.3. The 1 BSCFD Gas Hub Opportunity

The schematics for the 1 BSCFD gas gathering for the gas hub opportunity is shown below.

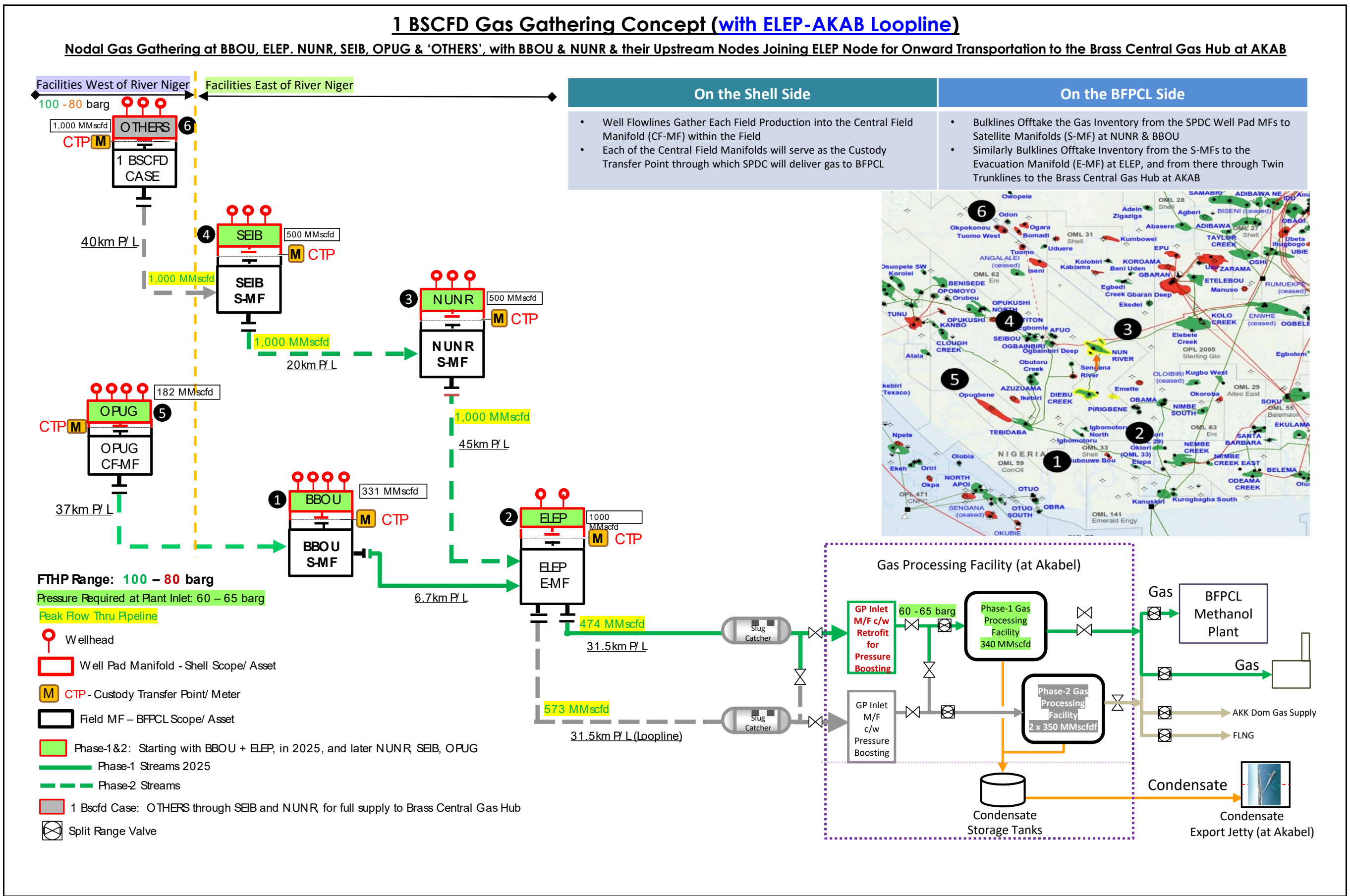


Figure 1.4 Schematics for the Gas Gathering of the1 BSCFD Gas Hub

Production from 'Other Node' flowing through SEIB & NUNR Nodes to ELEP Evacuation Manifold, and thence to AKAB Gas Hub, through the existing phases-1&2 ELEP – AKAB Pipeline, and a new Pipeline as a loop line to the existing ELEP - AKAB Pipeline, given that the phases 1&2 facilities are already in place.

The basic principle behind the details of the configuration is the optimal mix of line sizes and compression requirement to deliver the gas at the required pressure at destination.

1.4. Purpose of Document

The purpose of the document is to provide a structured summary of the key features of the FEED, - notably the project specifications, and specific deliverables required by regulators, - which collectively forms the technical and statutory basis for the project's Financial Close, and the subsequent EPC processes implementation.

2. Battery Limit, and Scope of Work

2.1. Battery Limit

The phase 1 facilities are the focus of the present FEED, however with consideration for provisions and retrofits that will accommodate the phase 2 facilities in a rather seamless manner.

The Phase-1 main facilities are:

- BBOU – ELEP pipeline
- ELEP - AKAB Pipeline
- BBOU Satellite Manifold
- ELEP Evacuation Manifold, and
- The Pipeline End Manifold at AKAB, complete with Pig Receiver.

The battery limits are shown below:

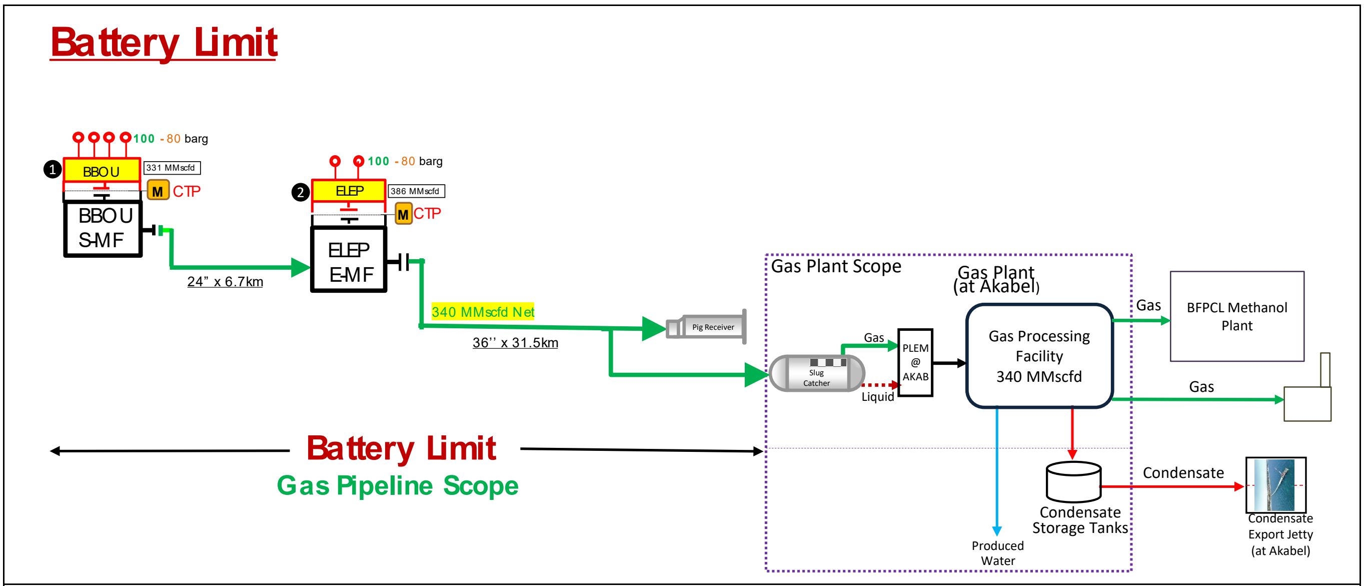


Figure 2.1 Battery Limit – Phase 1 Facilities

BBOU Platform

- The terminal flanged ball valve on the manifold collecting well fluids from BBOU Wells.
- The flanged ball valve provided for the tie-in of Opugbene (OPUG) Pipeline on the manifold transporting well fluids to ELEP.

ELEP Platform

- The terminal flanged ball valve on the manifold collecting well fluids from ELEPA Wells.
- The flanged ball valve provided for the tie-in of NUN River Pipeline on the manifold transporting well fluids to AKAB.

Akabel Facility

- The Pipeline terminal flange at the inlet to the slug catcher at AKAB.
- The drain outlet flange on the pig receiver at AKAB.

2.2. Scope of Work

The FEED scope covers, but not limited to the following:

1. Project Engineering
2. Process Systems Design, complete with Pipe Flow Hydraulics Analysis and Flow Assurance
3. Materials Specification
4. Pipeline Engineering
5. Mechanical and Piping Design
6. Structural Design
7. Civil and Foundation Engineering
8. Electrical Engineering
9. Control & Instrumentation, and Telecommunications Engineering
10. Technical Safety and Loss Prevention
11. EPC Cost Estimate

The Facilities to be covered under project scope are:

- Satellite Manifold at BBOU complete with all Process, Utility & Control Infrastructure, however with provision for future installation of PIG receiver as will be required for phase 2 production from OPUG.
- Evacuation Manifold at ELEP complete with all Process, Utility & Control Infrastructure, however with provision for future installation of PIG receiver as will be required for phase 2 production from NUNR.
- Pipeline from BBOU to ELEP
- Pipeline from ELEP to AKAB (up to Inlet Flange of Slug Catcher)
- PIG Launchers at BBOU and ELEP Manifold
- PIG Receivers at ELEP and AKAB
- Manifold control system also to be provided. Control Room required at each Manifold to house the Manifold Control system & Telecom system
- Cabinet Room to house the electrical MCC.
- UPS Power for Control System, associated Civil, Mechanical, Pipeline, Electrical, Instrumentation & Control, Telecommunication, and all other supporting infrastructure.

3. Project Engineering

3.1. Facilities Highlight

- API 5L X70 Mild Carbon Steel, to be concrete coated throughout the entire pipeline length for negative buoyancy (given the all-year flooded swampy terrain), and enhanced physical protection; except at river crossings where HDD method will be used.

  However, the continuous HDD (Horizontal Directional Drilling) option was evaluated against the backdrop of pipeline vandalization, and may be considered if the security risk assessment dictates as we progress.

- Manifolds integration between the BFPCL Pipeline Manifold Platforms and SPDC Well Pad Manifold Platforms at BBOU and ELEP.

- Submarine Armoured Power Cable from the Plant Complex at AKAB to ELEP and BBOU Manifolds for power supply to the manifolds.

3.2. Interfaces

The interfaces related to the project that need to be managed are as follows:

a. Interface of BFPCL Pipeline Manifolds with SPDC Well Pad Manifolds at BBOU and ELEP, in respect of power supply from BFPCL, tie-ins to the SPDC bulklines, metering, chemical injection, data transfer, telecommunications, utilities, drain systems, cold vent, control systems, safeguarding systems, operations start-up.

b. Interface of BFPCL Pipeline End Manifold at AKAB with BFPCL Slug Catcher, and the Plant Complex, in respect of power supply from BFPCL Captive Power Plant, tie-in to the Slug Catcher, metering, chemical injection, data transfer, telecommunications, utilities, drain systems, flare, control, & safeguarding systems, pre-commissioning, operations start-up.

c. The contractor will manage interfaces with subcontractors within its scope.

d. Site installation with con-current operations will require a robust interface and IAP (integrated activity planning) with the operations team for detailed design, shutdown, tie-ins, PTW management, etc. The BFPCL / SPDC OR&A engineers and representatives from the Production Operations function will form part of the project team. These personnel will be required to be part of key review events during detailed design, construction, commissioning and start-ups.

e. The BFPCL SD team will manage interfaces with communities and local contractors across the Pipeline stretch, through MOU that will be put in place, and framework that will be provided by the Project SP plan.

f. Other interfaces with partners, discipline engineering, regulatory agencies, etc. will be managed by the PMT.

3.3. Project Cost Estimate

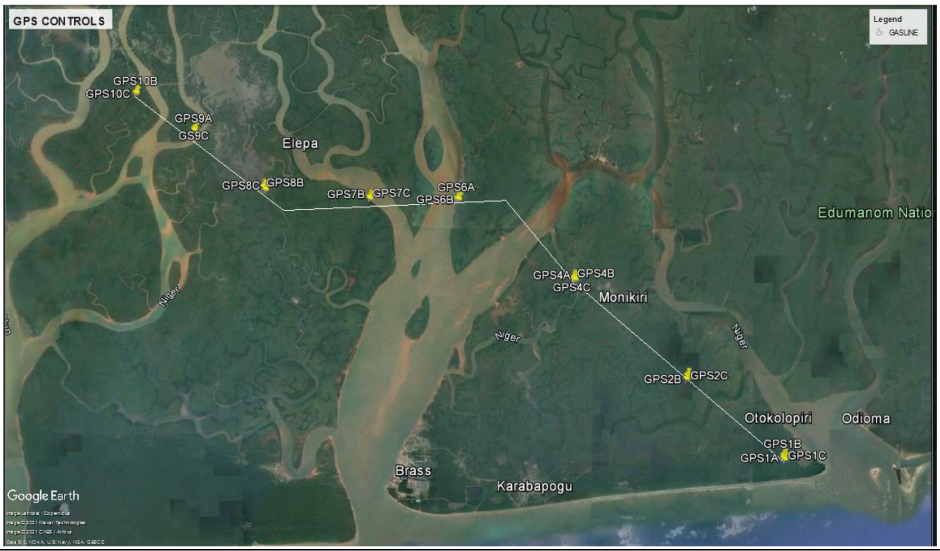
The EPC Cost Estimate of the Pipeline System is USD193,590,106.

S/N		Item Description	BOQ or Specification	Cost	
				NGN	USD
I	**Direct Cost**				
1	**Pipeline Cost**				
	1.1	Line Pipes	FEED Specification		
	1.2	Key Equipment	FEED Specification		
	1.3	Other Equipment	FEED Specification		
	1.4	Bulk Materials (Pipe / Valve / Flange / Miscellaneous)	FEED BOQ		
	1.5	Structural Steel	FEED BOQ		
	1.6	Cable & Electric	FEED BOQ		
	1.7	Civil Work	FEED BOQ		
	1.8	MEI Installation Work	FEED BOQ		
2	**Consumable for Initial Filling**				
3	**Incidental Cost of Eqpt & Materials**				
	3.1	Ocean Freight & Ocean Shipping Insurance			
	3.2	Local Transportation to Site			
II	**Indirect Cost**				
1	**Pre-Engineering Survey**				
2	**Detail Engineering Design**		Man-Hours Estimated		
3	**Project Management**		Man-Hours Estimated		
4	**Procurement Service**		Man-Hours Estimated		
5	**Construction Management**		Man-Hours Estimated		
6	**Insurance Fee**				
7	**Commissioning**				
8	**Permits Fee**				
9	**HSE Fee**				
10	**Training**				
	Total				

Ref Embed 3.2: EPC Cost Estimate of the Pipeline System	EPC Cost Estimate_EAL-05Jun2

3.4. The Phase 1 Pipelines Route / Right of Way

The pipeline route proposed for this phase 1 has been surveyed, and the survey report issued for the FEED.



The pipeline route / Right of Way, running from BBOU through ELEP to AKAB is shown in the satellite image on the left

Fig 3-1: Satellite Image of the proposed BBOU – ELEP – AKAB Right of Way

Ref: Fig 3-1: Satellite Image of the proposed BBOU – ELEP - AKAB Right of Way & Embed 3.4: BFPCL Bubouwe – Akabel Topographical	PDF BFPCL Bubouwe - Akabel Topographical

However, the route survey needs to be repeated as pre-eng survey to close the following gaps:

1. To incorporate pipeline route extension resulting from the current BBOU and ELEP Manifold locations advised by SPDC; and to follow the conventional description of from source of flow to destination, i.e., from BBOU to ELEP to AKAB.
2. To give proper offset from the ELEP Manifold. The present pipeline runs through the manifold location which should not be.
3. Rerouting of Pipeline approach to end at the slug catcher instead of the methanol plant.
4. Rerouting of pipeline route from Fantuo community (Bend B) to land at GPP, as dictated by political sensitivity around the Otokolopiri area makes it imperative to re-route pipeline

Ref Embed 3.4.2: Gas PL Route BBOU-ELEP-AKAB	PDF Gas PL Route BBOU-ELEP-AKAB

5. The parcellation survey did not precisely structure ownership communities on the RoW, more so as the new 2.5 km extension of the route is yet to be surveyed, and re-routing at the Akabel end is being considered, update to ownership of land has to be made.
6. The bathymetric data of some creeks and creeklets were not indicated on

the drawing.

7. All rivers, creeks and creek-lets should be labelled serially end-to-end with appropriate IDs for easy referencing.
8. Bathymetric chart datum should be tied to project elevation datum and not on predicted tide.
9. It's reported that topographical spot-heightening was conducted on ROW centreline at 25m interval, but the centreline was never traversed.
10. There is no topographical elevation data on the entire ROW

3.5. The Phase 1 Geotechnical Survey

The Geotechnical Survey has been done by Brone, and report issued for use in the foundation design.

Ref Embed 3.5: Soil Investigation Survey Draft Final Report	PDF Soil Investigation Survey_Draft Final Re

However, pre-Engineering Geotechnical survey for the pipeline Row also needs to be carried out to establish dense foundation strata, as well as cover the current pipeline manifold platform locations. In this respect, a geotechnical consultant and accredited soil testing laboratory will have to be engaged for analysis of data, - to enable close the following gaps:

1. Brone's thorough review of their recommendation to install piles at unstable soft formation of soil, which Epic does not see as feasible. This soft formation can be seen from the lithograph in the report prepared by BRONE.
2. EPIC position is that even at 15m depth, the soil is incapable of bearing manifold loads in the magnitude of more than 460Mt, hence need for further deeper soil investigation (This view is also shared by BRONE in its report).
3. From BRONE report, BH 20 that has the highest SPT - N counts are used by EPIC to compute the in-situ undrained shear strength and angles of internal friction of samples at various depths to 15m deepest termination layer. The estimated empirical result reveals a negative undrained shear strength; hence this layer of formation cannot support the imposed loads.

Ref Embed 3.4: Undrained Shear Strength Computation	UNDRAINED SHEAR STRENGTH COMPUT

4. The graphical illustration of the –ve undrain shear strengths/cohesion when plotted against SPT shows an inverse settlement of the pile when compared with +ve settlement depicting that at 15m the layer cannot settle uniformly (See embed file).

Ref Embed 3.5.4: Illustration of -ve Undrained Shear Strength	

5. There is no Geological model to show lateral and vertical extend of 'soil strata on the investigated locations, including the 'sand which may likely be silt' formation as depicted in BH 1,2,9 profile, hence the ''sand'' formation couldn't have just been profiled to terminate exactly at every 500 meters boundary as depicted in the lithology (see in BRONE report).

3.6. Design Criteria

3.6.1. Design Life

The Design Life of this project is 30 years.

3.6.2. Design Units

The FEED design was carried out using SI units. This is in accordance with DEP 00.00.20.10 – GEN (The use of SI Quantities and Units – Endorsement of ISO/IEC 80000) which specifies requirements and gives recommendations for the expression of quantities in SI units.

3.6.3. Cost Effectiveness & Life Cycle Cost

The Phase 1 & 2 pipelines are sized to deliver gas at the pressure required at Akabel, without the need for booster compressors, thereby avoiding higher Capex & Opex of compression when compared with the savings from use of smaller pipe size, coupled with opportunity cost owing to availability & reliability challenges associated with compressors.

3.6.4. Future Expansion - Phase-2 Facilities

The design has made provision for tie-ins at ELEP & BBOU Manifolds for the future Phase-2 production coming through NURN-ELEP & OPUG-BBOU Pipelines respectively.

3.6.5. Future Expansion – Proposed 1 Bscfd Gas Hub at Akabel

The Phase 1 & 2 pipelines are designed without the need for booster compression, and hence are at their maximum sizes. These pipeline sizes are therefore robust enough to support the addition of loopline to the ELEP-AKAB Pipeline in optimal mix with booster compressors, as future expansion in respect of realising the proposed 1 Bscfd Gas Hub at Akabel.

3.6.6. Buoyancy Combat

Concrete Coating through the entire pipeline length, for negative buoyancy, and enhanced physical protection, - except at River Crossings where there will be HDD (Horizontal Directional Drilling) across the river.

3.6.7. Flood Combat

Placement of the Pipeline Manifold Platforms at a height that is reasonably above flood level that will ever be experienced in the area, - drawing lessons from the 2012 rare flood experience.

3.6.8. Standardisation

Choice of wet gas metering at the Custody Transfer Point was standardised as Venturi Meter, which is what SPDC proposed to use at their Well Pad Manifold.

3.7. Local Requirements

3.7.1. Language

English language is the approved project language. All project documentation, drawings, correspondence were provided in English Language.

3.7.2. Local Content Requirement

The project shall ensure compliance with the Nigeria oil and gas industry content development act directives and associated requirement by the NCDMB (Nigerian content development monitoring board) The FEED of the project complied with the requirement for design with the FEED done in-country and minimum of 90% of the man-hours achieved in-country. The NCD law requires that:

- FEED and Detailed engineering on onshore facilities to be conducted in Nigeria with a minimum of 90% of the man hours in-country.
- Minimum fabrication of 80% by tonnage of steel in Nigeria or by registered OEM.
- Supply of the following materials where possible from Nigeria or by registered OEM: steel line pipes, steel plate, valves, LV and HV Cables etc.
- Employment and training of Nigerian labour and project personnel.

The project Nigerian content plan will be updated by the project management team before the start of detailed design.

3.8. Checklist of NUPRC Requirements

The checklist of NUPRC Requirements is summarised in Table 3.8 below.

Table 3.8: Checklist of NUPRC Requirements

S/N	Description	References
1	Project Quality Management Plan	PDF Project Quality Management Plan
2	Infrastructural Support Strategy & Manning	PDF Infrastr Support Strategy & Manning

3	Transient Simulation and Flow Assurance Report	Transient Simulation & Flow Assurance Rep
4	Stress Analysis Report	Piping Stress Analysis & Report-AK
5	Hazop Report	HAZOP Report
6	Inspection Plan for Line Pipes, Coating & Pipeline Fittings	Insp Plan for Linepipes, Coatn PL Fi
7	Pipeline Construction Plan inclusive of WPS / PQR.	
8	Pipeline Integrity Monitoring / Management Philosophy	Pipeline Intergrity Monitoring-Mgt Philo
9	Pipeline Blowdown & Depressurisation Plan	Pipeline Blowdown & Depressurisation Plan
10	Sewer System Planimetric Flow Diagram	Sewer System Planimetric Flow Diag
11	Decommissioning Plan	Decommissioning Plan
12	OPL Application Supporting Documents	OPL Application & Support Docs_Code-4
13	GIS Spatial Grid of the Pipeline System	
14	Regional Topography along Pipeline Route	
15	Schedule of Crossings	Schedule of Crossings-P1-1_Code-
16	Pipeline End Manifolds	

3.9. Checklist of OPL Application Requirements

The checklist of OPL Application Requirements is summarised in Table 3.9 below.

Table 3.9: Checklist of OPL Application Requirements

s/n	Description	References	
1	**PTS** PTS 3738 & 3746 in respect of BBOU-ELEP Pipeline & ELEP-AKAB Pipeline granted for a period of 2 years, with effect from 01 July 2022.	FRN PTS_3738_3746 Granted to BFPCL.pdf	

s/n	Description	References	
2	Service to be Rendered by Pipeline Transportation of Raw Natural Gas.	OPL Sketch-01.pptx	
3	Technical specification of the proposed Pipeline API 5L Grade X70.	OPL - Tech Spec of Pipeline.docx	
4	Characteristics of the fluid to be conveyed through the Pipeline Raw (Unprocessed) Wellhead Non-Associated Gas.	OPL - Fluid Characteristics.xlsx	
5	General description of the facility and its proposed design / operational parameters for any ancillary facilities along the pipeline or its terminal ends such as compressor stations, manifolds, meter banks etc	General Description.docx	OPL Sketch-01.pptx
6	Total estimated cost of construction $193.6 million, currently being updated	EPC Cost Estimate_EAL-05Jun2	
7	List of the prospective construction companies being considered for the construction 1. Don Mac Limited, Warri, Delta State 2. Enikkom Construction Ltd, Abuja, FCT 3. Esogra Engineering & Technical Services Limited Port Harcourt, Rivers State 4. MORPOL Engineering Services Ltd, Apapa, Lagos State		

Table 3.9: Checklist of OPL Application Requirements (contd.)

s/n	Description	References	
8	Detailed Survey Description of the Proposed Pipeline RoW in accordance with the Nigerian National Grid - showing width of the RoW with the coordinates (In approved geodetic formats) of the various points of intersection (10 copies) Will be updated during the pre-engineering survey.	PPL-DWG-001-R01_ Overall Location Pla	
9	Plan of the pipeline showing Proposed route of the pipeline marked in red, with sections and quarter sections shown Will be updated during the pre-engineering survey.	Gas Pipeline Route Survey_Gas PL Route	

s/n	Description	References	
10	Plan of the pipeline showing Locations of each point at which there is a change in any of the following parameters: a. Outside or Nominal Diameter of the Pipeline b. Wall thickness of the Linepipe Material c. Type & Grade of Linepipe d. Design / Maximum Operating Pressure a. Diameter changed at Elepa from 24” (for the BBOU-ELEP Segment) to 36” (for the ELEP-AKAB Segment) b. Wall thickness changed at Elepa from 15.88 mm (for the 24” x 6.7km for the BBOU-ELEP Segment) to 23.88 mm (for the 36” x 31.5km ELEP-AKAB Segment) c. No change in Type & Grade of Linepipe d. No change in Design / Maximum Operating Pressure	OPL Sketch-02.pptx	
11	Plan of the pipeline showing Direction of fluid flow along the pipeline From BBOU to ELEP, and From ELEP to AKAB	OPL Sketch-02.pptx	
12	Plan of the pipeline showing Locations indicated by Symbols of Installations along the Pipeline. a. Pipeline Manifold Platform at BBOU b. Pipeline Manifold Platform at ELEP c. Pipeline End Manifold at AKAB	3D MODEL - BBOU Manifold 3D Model - PLEM at AKAB	3D MODEL - ELEP Manifold

Table 3.9: Checklist of OPL Application Requirements (contd.)

s/n	Description	References	
13	Plan of the pipeline showing Locations of the points at which other pipelines would be crossed by the new pipeline indicating the owner of the pipeline(s) to be crossed. a. Crossing of 3 NAOC Pipelines, and b. Crossing of 1 SPDC Pipeline	Gas Pipeline Route Survey_Gas PL Route	PPL-DWG-006_Typical Pipeline Crossing
14	Plan of the pipeline showing Regional topography of the area along the pipeline route within a distance of 100 meters on both sides, including watercourses to be crossed Niger Delta Mangrove Swamp as shown on the survey map		
15	Plan of the pipeline showing Location of any anchors or expansion loops None		

s/n	Description	References	
16	Plan of the pipeline showing Location and operating details of corrosion prevention devices, main line valves and any emergency shut down devices CP System, Pig Launchers / Receivers, Mainline Valves, & Shutdown Valves are on the Manifolds Platform – shown on the PFD	PFDs BBOU, ELEP, & AKAB Manifold Platfo	
17	Plan of the pipeline showing Location and operating details of corrosion prevention devices, main line valves and any emergency shut down devices CP System, Pig Launchers / Receivers, Mainline Valves, & Shutdown Valves are on the Manifolds Platform – shown on the PFD	PFDs BBOU, ELEP, & AKAB Manifold Platfo	
18	Plan of the pipeline showing Pig launching and receiving points and any tie-in points with operating details Pig Launchers & Receivers installed on the Pipeline Manifold Platforms, and Pig Receiver installed on the Pipeline End Manifold at AKAB	P&IDs - BBOU, ELEP, & AKAB	
19	Hydraulic profile of the pipeline indicating the position of any pumping or booster stations and velocity profile (indicating erosional velocity boundaries) a. No Pumping or Booster Station b. Velocities are in region of 1 – 3 m/s, while the erosional velocities are above 11 m/s	OPL Sketch-03.pptx	

Table 3.9: Checklist of OPL Application Requirements (contd.)

s/n	Description	References	
20	Plan and profile of the pipeline showing the manner in which any water way or other pipeline lying along the route would be crossed Pig Launchers & Receivers installed on the Pipeline Manifold Platforms, and Pig Receiver installed on the Pipeline End Manifold at AKAB	PPL-DWG-007_Typical River Crossing_EA	PPL-DWG-006_Typical Pipeline Crossing_
21	Pipeline Crossing Agreement with NAOC Proforma Crossing Agreement ready and will be finalised at EPC stage.	Proforma Pipeline Crossing Agreement	
22	Pipeline Crossing Agreement with SPDC Yet to receive feedback from SPDC in respect of the status of the line and its new Owner.	Gas Pipeline Route Survey_Gas PL Route	
23	OPL Application TBA.		

4. Design Basis & Design Data

The FEED was based on 340 MMscfd Instantaneous NAG production, translated from DCQ 310 Forecast from SPDC.

The starting point for the facilities sizing was 386 MMscfd of gas, 34 Mbcd of condensate, and 16 Mpwd of water, which were maximum rates constructed by uprating DCQ 270 Total Nodal Forecasts to DCQ 310, pending receipt of the DCQ 310 MMscfd forecast from SPDC.

The DCQ 310 MMscfd forecast has now (Jun 2022) been received, which has been used to validate facilities sizing, and is in excellent agreement.

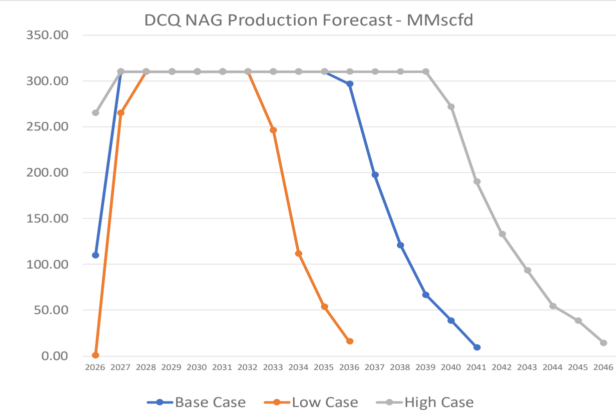
4.1. The DCQ 310 Production Forecast

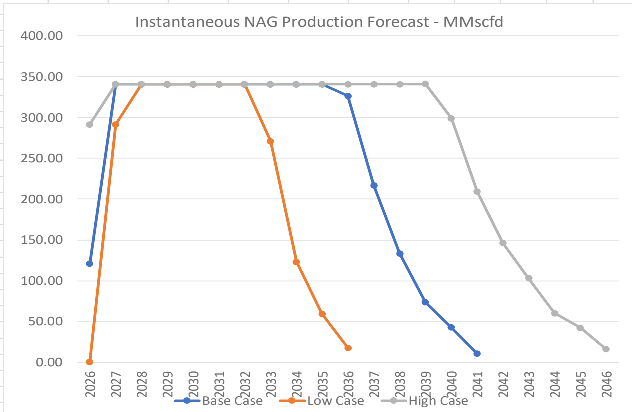
The DCQ NAG Production Forecast from which the Instantaneous NAG Forecast is shown below, along with the Water Production Forecast.

The DCQ forecast includes 10% low and off, which when added back results on Instantaneous Production Forecast which is 110% of the DCQ forecast.

Gross DCQ NAG Production Forecast (MMscfd)

	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039	2040	2041	2042	2043	2044	2045	2046
Base Case	109.77	310.00	310.00	310.00	310.00	310.00	310.00	310.00	310.01	310.03	296.74	197.31	121.16	66.78	38.90	9.50					
Low Case	0.63	264.99	310.00	310.00	310.00	310.01	310.00	246.41	111.79	53.94	15.96										
High Case	264.99	310.00	310.00	310.00	310.00	310.00	310.00	310.00	310.02	310.04	310.04	310.04	310.03	310.04	271.88	190.03	133.08	93.64	54.55	38.43	14.44





Gross Instantaneous NAG Production Forecast (MMscfd)

	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039	2040	2041	2042	2043	2044	2045	2046
Base Case	120.75	341.00	341.00	341.00	341.00	341.00	341.00	341.00	341.01	341.03	326.41	217.04	133.27	73.46	42.79	10.45					
Low Case	0.69	291.49	341.00	341.00	341.00	341.01	341.00	271.05	122.97	59.34	17.55										
High Case	291.49	341.00	341.00	341.00	341.00	341.00	341.00	341.00	341.02	341.04	341.04	341.04	341.04	341.04	299.07	209.03	146.38	103.00	60.01	42.27	15.88

Figure 4-1: DCQ 310 Production Profile and Instantaneous Production Profile

4.2. Water Production Forecast

Year	BFGS Phase-1 Water Prod Forecast (BwpD)						Liquid Handling Capacity B/D
	Plant Forecast	BB-D3 well-01	BB-D3 well-02	BB-D3 well-03	ELEPA U4	ELEPA U7	
01/01/2026	0	0	0	0	0	0	20000
01/01/2027	0	0	0	0	0	0	20000
01/01/2028	2	1	1	1	0	0	20000
01/01/2029	12	4	4	4	0	1	20000
01/01/2030	44	13	13	13	1	3	20000
01/01/2031	142	45	45	45	3	7	20000
01/01/2032	394	124	124	124	8	17	20000
01/01/2033	1094	347	347	347	18	40	20000
01/01/2034	3368	1083	1083	1083	39	85	20000
01/01/2035	11104	3615	3615	3615	80	178	20000
01/01/2036	464	0	0	0	164	300	20000
01/01/2037	638	0	0	0	337	300	20000
01/01/2038	1003	0	0	0	703	300	20000
01/01/2039	1825	0	0	0	1525	300	20000
01/01/2040	3852	0	0	0	3552	300	20000
01/01/2041	9739	0	0	0	9439	300	20000
01/01/2042	32248	0	0	0	31948	300	20000
01/01/2043	32248	0	0	0	31948	300	20000

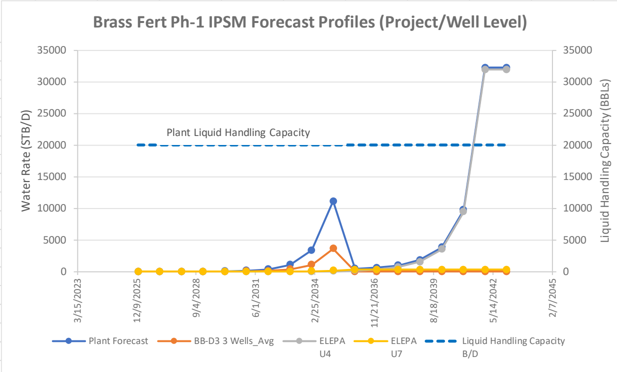


Figure 4-2: Total Liquid Production Profile

4.3. The Flow Properties & Fluid Composition

The fluid properties and compositions used for the hydraulics analysis are as shown below. They are analogue compositions, with BB-D3 analogue assumed for BBOU and OPUG, and ELEP assumed for ELEP, NUNR, and SEIB.

BB-D3 Reservoir Fluid Composition & Properties

BB-D3 Reservoir Fluid Composition

No.	Component	Flash Gas [mol %]	Flash Oil [mol %]	Res Fluid [mol %]
1	N2	0.07	0.00	0.07
2	CO2	0.59	0.00	0.59
3	C1	93.26	0.00	92.92
4	C2	3.17	0.01	3.16
5	C3	1.01	0.03	1.01
6	i-C4	0.61	0.07	0.61
7	n-C4	0.44	0.07	0.44
8	i-C5	0.25	0.16	0.25
9	n-C5	0.19	0.16	0.19
10	C6	0.22	1.88	0.23
11	C7	0.17	6.21	0.19
12	C8	0.02	6.61	0.04
13	C9	0.00	7.13	0.03
14	C10	0.00	5.70	0.02
15	C11	0.00	7.83	0.03
16	C12+	0.00	64.14	0.22
	TOTAL	**100**	**100**	**100**

BB-D3 Reservoir Fluid Properties

		Flash Gas	Flash Oil	Res Fluid
Density @ 60	[g/cm^3]		0.811	
MW	[g/mol]	17.97	165.61	18.47
GOR	[Scf/Stb]			180,012
Gravity	[air=1]	0.62		
C_{7+}	Mole %	0.45	97.62	0.53
C_{7+} MW	[g/mol]	100.89	167.69	143.17
C_{12+} MW	[g/mol]	-	191.6	191.6
CGR (bbls/mmscf]				5.56

ELEP U4/U7 Reservoir Fluid Composition & Properties

ELEP U4/U7 averaged Res Fluid Composition

No.	Component	Flash Gas [mol %]	Flash Oil [mol %]	Res Fluid [mol %]
1	N2	1.08	0.00	1.07
2	CO2	1.15	0.00	1.11
3	C1	85.97	0.00	84.17
4	C2	6.30	0.20	6.17
5	C3	2.37	0.26	2.32
6	i-C4	0.64	0.27	0.64
7	n-C4	0.66	0.40	0.65
8	i-C5	0.42	0.63	0.35
9	n-C5	0.16	0.52	0.25
10	C6	0.43	2.07	0.47
11	C7	0.51	6.36	0.64
12	C8	0.33	13.14	0.59
13	C9	0.00	12.10	0.24
14	C10	0.00	9.33	0.19
15	C11	0.00	7.05	0.14
16	C12	0.00	6.45	0.14
17	C13	0.00	6.58	0.14
18	C14	0.00	6.09	0.13
	C15	0.00	5.25	0.12
	C16	0.00	4.20	0.09
	C17	0.00	3.18	0.07
	C18	0.00	2.93	0.07
	C19	0.00	1.93	0.04
	C20+	0.00	11.10	0.25
	TOTAL	**100.00**	**100.00**	**100.01**

ELEP U4 Reservoir Fluid Properties

		Flash Gas	Flash Oil	Res Fluid
Density @ 60	[g/cm^3]		0.827	
MW	[g/mol]	20.84	181.08	26.44
GOR	[Scf/Stb]			
Gravity	[air=1]	0.7190		0.9130
C_{7+}	Mole %	1.13	94.98	4.41
C_{7+} MW	[g/mol]	105.67	186.72	166.70
$C20_{+}$ MW	[g/mol]	-	339.50	339.50
CGR (bbls/mmscf]				

ELEP U7 Reservoir Fluid Properties

		Flash Gas	Flash Oil	Res Fluid
Density @ 60	[g/cm^3]		0.839	
MW	[g/mol]	19.43	175.14	20.55
GOR	[Scf/Stb]			
Gravity	[air=1]	0.6710		0.7090
C_{7+}	Mole %	0.54	96.34	1.24
C_{7+} MW	[g/mol]	105.66	179.07	146.96
$C20_{+}$ MW	[g/mol]	-	338.80	338.80
CGR (bbls/mmscf]				

4.4. The Nodal Production Forecast

	BBOU TO ELEPA PIPELINE SEGMENT (24" x 6.7km)								
	BBOU NODE								
	TOTAL FLUID RATE			BUBOUWE BOU			OPUGBENE		
year	MMscfd	bcpd	bwpd	MMscfd	bcpd	bwpd	MMscfd	bcpd	bwpd
0	(gas)	(cond.)	(water)	(gas)	(cond.)	(water)	(gas)	(cond.)	(water)
1	181	944	907	181	944	907	0	0	0
2	181	878	907	181	878	907	0	0	0
3	181	788	907	181	788	907	0	0	0
4	181	742	7,855	181	742	7,855	0	0	0
5	181	707	7,547	181	707	7,547	0	0	0
6	215	840	8,491	215	840	8,491	0	0	0
7	305	1,217	9,002	305	1,217	9,002	0	0	0
8	327	1,323	8,193	327	1,323	8,193	0	0	0
9	331	1,338	8,038	331	1,338	8,038	0	0	0
10	93	1,763	2,527	67	239	2,397	26	1,525	129
11	75	3,242	1,301	21	59	1,031	54	3,183	270
12	74	3,457	1,074	16	42	784	58	3,416	290
13	71	4,084	1,513	2	4	85	69	4,080	1,429
14	71	4,161	3,048	0	0	0	71	4,161	3,048
15	94	5,541	3,748	0	0	0	94	5,541	3,748
16	154	9,082	3,859	0	0	0	154	9,082	3,859
17	170	12,115	3,148	0	0	0	170	12,115	3,148
18	174	16,226	1,667	0	0	0	174	16,226	1,667
19	177	15,812	1,558	0	0	0	177	15,812	1,558
20	182	15,606	2,775	0	0	0	182	15,606	2,775

	ELEPA TO AKABEL PIPELINE SEGMENT (36" x 31.5km)																	
	ELEP NODE																	
year	TOTAL FLUID RATE			BUBOUWE BOU			ELEPA			NUN RIVER			SEIB			OPUG		
	MMscfd	bcpd	bwpd	MMscfd	bcpd	bwpd	MMscfd	bcpd	bwpd	MMscfd	bcpd	bwpd	MMscfd	bcpd	bwpd	MMscfd	bcpd	bwpd
	(gas)	(cond.)	(water)	(gas)	(cond.)	(water)	(gas)	(cond.)	(water)	(gas)	(cond.)	(water)	(gas)	(cond.)	(water)	(gas)	(cond.)	(water)
1	361	11,032	1,807	181	944	907	180	10,088	900	0	0	0	0	0	0	0	0	0
2	361	10,965	1,807	181	878	907	180	10,088	900	0	0	0	0	0	0	0	0	0
3	361	10,876	1,807	181	788	907	180	10,088	900	0	0	0	0	0	0	0	0	0
4	361	10,830	15,835	181	742	7,855	180	10,088	7,980	0	0	0	0	0	0	0	0	0
5	361	10,794	15,213	181	707	7,547	180	10,088	7,666	0	0	0	0	0	0	0	0	0
6	360	8,985	13,393	215	840	8,491	145	8,146	4,902	0	0	0	0	0	0	0	0	0
7	357	4,128	11,599	305	1,217	9,002	52	2,911	2,597	0	0	0	0	0	0	0	0	0
8	356	2,953	9,647	327	1,323	8,193	29	1,630	1,454	0	0	0	0	0	0	0	0	0
9	356	2,724	9,275	331	1,338	8,038	25	1,386	1,236	0	0	0	0	0	0	0	0	0
10	382	18,828	4,837	67	239	2,397	19	1,076	960	244	14,464	1,221	26	1,525	129	26	1,525	129
11	385	21,637	3,357	21	59	1,031	11	628	561	245	14,583	1,225	54	3,183	270	54	3,183	270
12	385	21,933	3,015	16	42	784	9	478	426	245	14,583	1,225	58	3,416	290	58	3,416	290
13	386	22,799	13,832	2	4	85	1	52	46	245	14,583	10,844	69	4,080	1,429	69	4,080	1,429
14	386	22,905	16,513	0	0	0	0	0	0	245	14,583	10,417	71	4,161	3,048	71	4,161	3,048
15	384	21,996	14,149	0	0	0	0	0	0	196	10,914	6,653	94	5,541	3,748	94	5,541	3,748
16	379	22,323	11,246	0	0	0	0	0	0	71	4,159	3,528	154	9,082	3,859	154	9,082	3,859
17	380	26,558	8,272	0	0	0	0	0	0	40	2,329	1,976	170	12,115	3,148	170	12,115	3,148
18	382	34,433	5,014	0	0	0	0	0	0	34	1,980	1,680	174	16,226	1,667	174	16,226	1,667
19	380	33,161	4,421	0	0	0	0	0	0	26	1,538	1,304	177	15,812	1,558	177	15,812	1,558
20	379	32,111	6,313	0	0	0	0	0	0	15	898	762	182	15,606	2,775	182	15,606	2,775

5. Process Systems Design

The following are specific key drivers to the Process Systems Design:

1. Verification & Validation of Pipeline Size – Phase 1 Facilities
2. Pipeline Network & Pipe Sizing that will deliver the gas at the required pressure without the need for booster compressors. This scenario is more cost effective and operationally more convenient, considering life cycle cost (CAPEX + OPEX), utilities, operations, and maintenance requirement of compressors, as well as reliability / availability issues associated with rotating equipment.
3. Flow Assurance
4. Process Design (PFS & PEFS) with sound process control & safeguarding capabilities

5.1. Validation of Pipeline Size – Phase 1 Facilities

The results speak to the case of 36" Pipeline for ELEP – AKAB, - meeting criteria at both THPs.

Field Manifolds & End Manifold Pressures

Case-1 Nodes THP of 100 barg

		Segment	Pipe ID	Single Phase Gas Flow: Pressures (barg) BBOU M/F	ELEP M/F	AKAB	Pressure Drop barg	barg / km	Two Phase Gas Flow: Pressures (barg) BBOU M/F	ELEP M/F	AKAB	Pressure Drop barg	barg / km
		24" x 6.7 km BBOU - ELEP	22.77"	100.00	98.01		1.99	0.30	100.00	97.77		2.23	0.33
24" x 6.7 km BBOU - ELEP	with	36" x 31.5 km ELEP - AKAB	34.15"		98.01	88.63	9.38	0.30		97.77	88.08	9.69	0.31
24" x 6.7 km BBOU - ELEP	with	32" x 31.5 km ELEP - AKAB	30.60"		98.01	82.16	15.85	0.50		97.77	81.59	16.19	0.51
24" x 6.7 km BBOU - ELEP	with	28" x 31.5 km ELEP - AKAB	26.72"		98.01	63.62	34.39	1.09		97.77	63.01	34.77	1.10
24" x 6.7 km BBOU - ELEP	with	24" x 31.5 km ELEP - AKAB	22.77"		98.01	43.18	54.83	1.74		97.77	42.68	55.10	1.75
		Pressure Required (barg)				60 - 65					60 - 65		

Case-2 Nodes THP of 80 barg

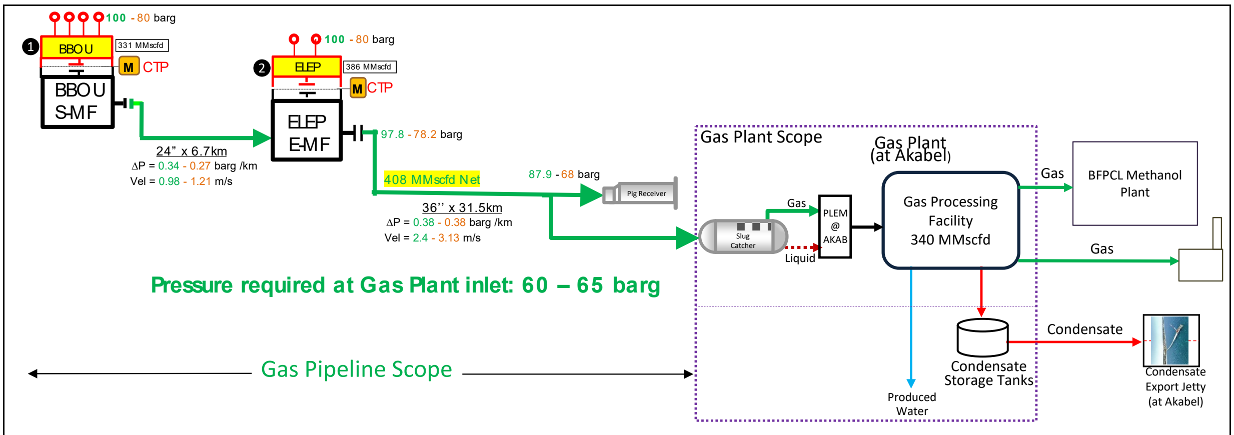
		Segment	Pipe ID	Single Phase Gas Flow: Pressures (barg) BBOU M/F	ELEP M/F	AKAB	Pressure Drop barg	barg / km	Two Phase Gas Flow: Pressures (barg) BBOU M/F	ELEP M/F	AKAB	Pressure Drop barg	barg / km
		24" x 6.7 km BBOU - ELEP	22.77"	80.00	78.34		1.66	0.25	80.00	78.00		2.00	0.30
24" x 6.7 km BBOU - ELEP	with	36" x 31.5 km ELEP - AKAB	34.15"		78.34	68.91	9.42	0.30		78.00	68.17	9.83	0.31
24" x 6.7 km BBOU - ELEP	with	32" x 31.5 km ELEP - AKAB	30.60"		78.34	62.67	15.66	0.50		78.00	62.14	15.86	0.50
24" x 6.7 km BBOU - ELEP	with	28" x 31.5 km ELEP - AKAB	26.72"		78.34	42.36	35.98	1.14		78.00	41.73	36.27	1.15
24" x 6.7 km BBOU - ELEP	with	24" x 31.5 km ELEP - AKAB	22.77"		78.34	23.42	54.91	1.74		78.00	22.85	55.16	1.75

Legend
- Line Size Meets Requirement with reasonable margin
- Borderline Case with No Margin
- Does Not Meet Requirement

5.2. Pipeline Network Simulation with Different Pipeline Sizes

Phase-1 Facilities – BBOU & ELEP

The result validates 24" for BBOU-ELEP, and 36" for ELEP-AKAB as shown below, satisfying the band of FTHPs.



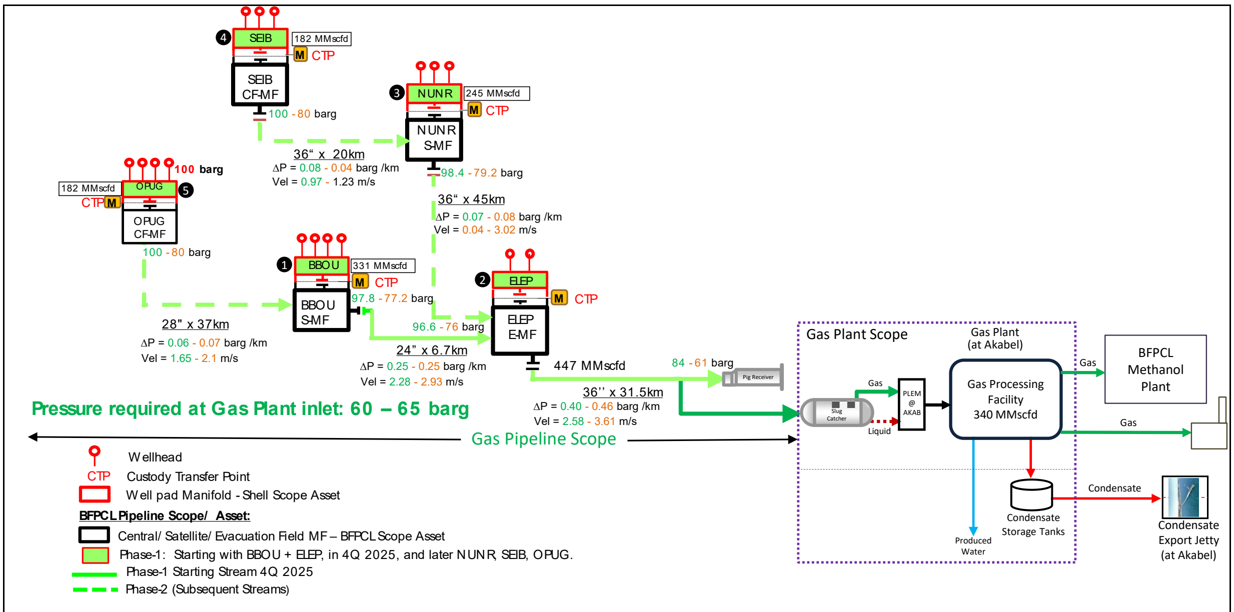
Phase-1 : PipeFlow Parameters: BBOU - ELEP - AKAB

Phase	Section	S/N	Pipeline Size BBOU-ELEP (in)	Pipeline Size ELEP-AKAB (in)	FTHP at Start Nodes BBOU & ELEP (barg)	Delivered Pressure ELEP M/F (barg)	Delivered Pressure AKAB (barg)	Pressure Drop (barg)	Pressure Drop (barg / km)	Velocity (ft / sec)	Velocity (m / sec)	Erosional Velocity (ft / sec)	Erosional Velocity (m / sec)	Erosional Velocity Ratio (EVR)	Remarks
Phase - 1	BBOU - ELEP	1	16	24	100	97.19		2.81	0.42	45.94	14.00	43.01	13.11	1.068	
					80	76.77		3.23	0.48	69.95	21.32	53.08	16.18	1.318	
		2	20	24	100	98.76		1.24	0.19	45.94	14.00	43.01	13.11	1.068	
					80	77.97		2.03	0.30	69.95	21.32	53.08	16.18	1.318	
		3	24	24	100	98.63		1.37	0.20	3.21	0.98	28.80	8.78	0.112	
					80	78.52		1.48	0.22	4.01	1.22	32.16	9.80	0.125	
		4	24	28	100	97.75		2.25	0.34	3.24	0.99	28.93	8.82	0.112	
					80	78.19		1.81	0.27	4.05	1.23	32.40	9.88	0.125	
		5	24	36	100	97.75		2.25	0.34	3.22	0.98	28.71	8.75	0.112	
					80	78.19		1.81	0.27	4.02	1.23	32.16	9.80	0.125	
	ELEP - AKAB	1	16	24	100	97.19	75.00	25.00	0.79	20.78	6.33	37.64	11.47	0.552	Meets Requirement
					80	76.77	46.00	34.00	1.08	34.46	10.50	48.47	14.77	0.711	Insufficient Pressure at AKAB
		2	20	24	100	98.76	77.00	23.00	0.73	20.22	6.16	37.17	11.33	0.544	Meets Requirement
					80	77.97	48.00	32.00	1.02	32.98	10.05	47.45	14.46	0.695	Insufficient Pressure at AKAB
		3	24	24	100	98.63	76.30	23.70	0.75	20.68	6.30	37.33	11.38	0.554	Meets Requirement
					80	78.52	48.00	32.00	1.02	33.42	10.19	47.45	14.46	0.704	Insufficient Pressure at AKAB
		4	24	28	100	97.75	84.66	15.34	0.49	13.39	4.08	35.42	10.80	0.378	Meets Requirement
					80	78.19	62.92	17.08	0.54	18.30	5.58	41.59	12.68	0.440	Meeting Requirement with Insufficient Margin
		5	24	36	100	97.75	87.92	12.08	0.38	7.85	2.39	34.73	10.59	0.226	Meets Requirement
					80	78.19	68.02	11.98	0.38	10.28	3.13	39.69	12.10	0.259	Meets Requirement
	MAX TRUPUT	1	24	36	100	87.16	65	35.00	1.11	33.34	10.16	48.56	14.80	0.687	
					80	77.09	65	15.00	0.48	17.90	5.45	44.94	13.70	0.398	
	MIN TRUPUT	2	24	36	100	97.74	65	35.00	1.11	3.47	1.06	27.86	8.49	0.125	
					80	78.07	65	15.00	0.48	3.34	1.02	28.64	8.73	0.117	

	BBOU	ELEPA	Total Max Gas MMSCFD	
MAX TRUPUT (100)	945	316	1,261	Based on Min Pressure Reqd at AKAB
MAX TRUPUT (80)	340	316	656	Based on Min Pressure Reqd at AKAB
MIN TRUPUT (100)	70	85	155	Based on Min Vel for Pipe Cleaning
MIN TRUPUT (80)	50	50	100	Based on Min Vel for Pipe Cleaning

Phase-2 Facilities – BBOU + ELEP + NUNR + SEIB + OPUG

The result validates 24” for BBOU-ELEP, and 36” for ELEP-AKAB as shown below, satisfying the band of FTHPs.



Phase-2: PipeFlow Hydraulics Parameters: Total Nodes Flow to AKAB thru ELEP

			Pipeline Size OPUG-BBOU	Pipeline Size BBOU-ELEP	Pipeline Size SEIB-NUNR	Pipeline Size NUNR-ELEP	Pipeline Size ELEP-AKAB	FTHP at Start Nodes OPUG & SEIB	Delivered Pressure BBOU M/F	Delivered Pressure NUNR M/F	Delivered Pressure ELEP M/F	Delivered Pressure AKAB	Pressure Drop (Segment ELEP - AKAB)		Velocity (Segment ELEP - AKAB)		Erosional Velocity (Segment ELEP - AKAB)		Erosional Velocity Ratio (EVR)	
			in	in	in	in	in	barg	barg	barg	barg	barg	barg	barg / km	ft / sec	m / sec	ft / sec	m / sec		
Phase-2	Segment OPUG - BBOU - ELEP	1	**20**	20	36	36	24	**100**	86.43		83.05									
								80	62.14		56.66									
		2	**24**	24	36	36	24	**100**	95.11		93.56									
								80	74.13		72.26									
		3	**28**	24	36	36	28	**100**	97.84		96.18									
								80	77.23		75.55									
		4	**28**	24	36	36	32	**100**	98.54		97.02									
								80	77.40		75.59									
		5	**28**	24	36	36	36	**100**	98.14		96.54									
								80	78.46		76.10									
		6	**24**	24	36	36	36	**100**	95.17		93.55									
								80	74.14		72.39									
	Segment SEIB - NUNR - ELEP	1	28	24	**28**	**28**	24	**100**		99.66	91.74									
								87		85.78	78.22									
		2	28	24	**36**	**36**	24	**100**		98.65	95.53									
								82		80.4	78.16									
		3	28	24	**36**	**36**	28	**100**		99.22	95.47									
								80		79	75.55									
		4	28	24	**36**	**36**	32	**100**		99.59	96.46									
								80		79.17	74.04									
		5	28	24	**36**	**36**	36	**100**		98.42	96.56									
								80		79.17	75.55									
	Segment ELEP - AKAB	1	28	24	36	36	**24**	**100**			95.53	55.00	40.53	1.29	22.17	6.76	43.47	13.25	0.51	
								82			78.16	5.00	73.16	2.32	221.23	67.43	137.24	41.83	1.612	
		2	28	24	36	36	**28**	**100**			95.47	67.00	28.47	0.90	17.60	5.36	41.61	12.68	0.423	Meets Requirement
								80			75.55	35.00	40.55	1.29	34.70	10.58	58.32	17.78	0.595	Insufficient Pressure at AKAB
		3	28	24	36	36	**32**	**100**			96.46	78.00	18.46	0.59	12.04	3.67	38.47	11.72	0.313	Meets Requirement
								80			74.04	51.00	23.04	0.73	18.95	5.78	48.22	14.70	0.393	Insufficient Pressure at AKAB
		4	28	24	36	36	**36**	**100**			96.56	84.00	12.56	0.40	8.48	2.58	37.03	11.29	0.229	Meets Requirement
								80			75.55	61.00	14.55	0.46	11.85	3.61	43.73	13.33	0.271	Insufficient Margin
		5	24	24	36	36	**36**	**100**			96.56	84.00	12.56	0.40	8.48	2.58	37.03	11.29	0.229	Meets Requirement
								80			75.55	61.00	14.55	0.46	11.85	3.61	43.73	13.33	0.271	Insufficient Margin

5.3.

5.4. Flow Assurance

The following were considered critical to the flow assurance:

1. Capability to get to destination at the required pressure.
2. The absence of conditions that will impair or obstruct flow, - hydrate formation, wax, asphaltene, and scale deposition, emulsion formation, and slugging.

The flow assurance analysis of the Multiphase Production Pipelines from BBOU to Akabel comprises the following:

- Steady State Analysis – to determine operating conditions over field life, confirm/validate the pipeline size against the steady state hydraulic analysis results.
- Slugging Assessment – to determine the impact of the line size selected on slugging and pipeline operability by reporting the flow regime and the instantaneous liquid arrival at Akabel.
- Pigging Analysis – to determine the liquid surge volume as a result of pigging operation to aid the design of slug catcher.
- Cooldown and Pipeline Depressurisation Analysis – to determine the cooldown rate and assess any risk of hydrate when the pipeline is shut-in for extended periods as well as during depressurisation of the pipeline:
  - Liquid arrival volume at Akabel
  - Minimum fluid temperature downstream of the RO.
  - Peak gas flowrate
- Hydrate Risk Assessment – to assess the risk of wax or hydrate formation during steady state and transient operation.

5.4.1. Steady State

- The 24” x 36” pipeline are optimal for delivering the production forecast to Akabel based on the required pressure being within the available pressure range.
- The 24” x 36” pipeline shows no fluctuation in the pipeline pressure which is representative of stability.
- The erosional velocity limit based on API14E is not exceeded with the 24” x 36” "pipeline so no erosional velocity constraint exists with this line sizes.

5.4.2. Slugging Assessment

- Examination of the flow regimes in the pipelines show that flow is mostly stratified along the pipeline except at low point where it is slug flow.
- Instantaneous liquid arrival flowrates at Akabel however show significant fluctuations for the 24” x 36” pipeline. This translates into significant levels of fluctuation in liquid arrival at Akabel. In contrast, stable liquid arrival flowrate is obtained for the 24” x 28” pipeline, however, the 28” pipeline does not meet requirement at 80 barg FTHP.

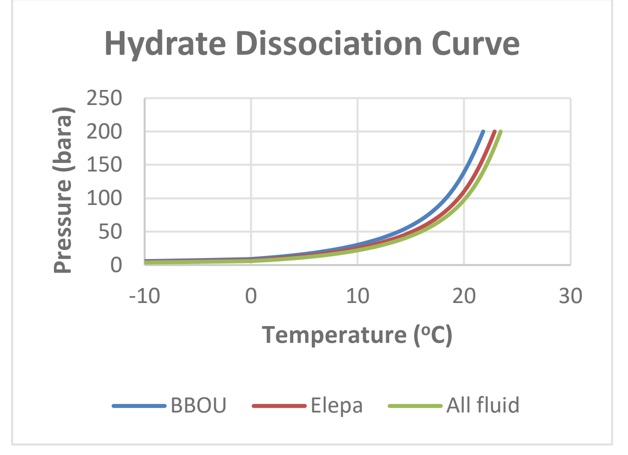
5.4.3. Turndown Assessment

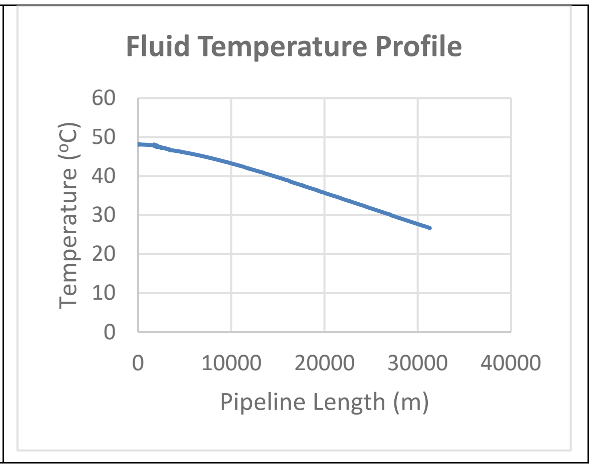
- When the total flowrate is lower than 150 MMscfd the system is gravity dominated while when the total flowrate is higher than 150 MMscfd the system is friction dominated.

- The fluid velocity for flowrates lower than 150 MMscfd is less than the minimum specified by API 14E (flow velocity should not be less than 3 ft/s (1 m/s) to minimize deposition of sand and other solids).

5.4.4. Hydrate Assessment

- At the minimum swamp temperature of 18°C, there is no risk of hydrates forming in the pipeline during steady state operation as the steady state temperature is above the hydrate formation temperature at the operation pressure of 74 bara. Also, at the settle-out pressure of 66 bara during pipeline shutdown, it is concluded that there is no risk of hydrate formation in the pipeline during shutdown conditions.





5.4.5. Cooldown & Pipeline Depressurization Results

- The pipeline temperature profile during steady state operation shows the fluid temperature is above the hydrate formation temperature at the pipeline outlet/arrival at Akabel. Hence cooling of the pipeline following shutdown poses no additional hydrates risk.
- On pipeline shutdown, the pipeline settles-out at approximately 66 bara. At this pressure and ambient temperature (minimum swamp temperature of 18°C) no hydrate risk exists.
- The rate of pipeline depressurization is dependent on the RO size. The optimum RO size however cannot be determined as further analysis to include the pipework downstream of the RO should be considered and the impact of the depressurization rate, RO size and subsequent Joule-Thompson cooling on minimum wall temperature assessed to inform the RO size. This is recommended to be done at the detailed engineering design stage.
- The lowest temperature obtained downstream the RO (4-inch) is -16.9°C during pipeline depressurization.
- The water outflows from the pipeline during depressurization is negligible for the RO sizes considered.
- The oil outflow from the pipeline increases with RO size increase amounting to 487m3 for RO size of 4-inch.
- The maximum/peak instantaneous gas flowrate of 334 MMscfd is obtained during pipeline depressurization when an RO of 4 inch is used.

5.4.6. Pigging Results

- The pig residence time when travelling at an average velocity of 1 and 1.5 m/s is approximately 8 and 5.5 hours respectively.
- During pigging operation, a maximum accumulated volume of 3500 m^3 is calculated.

5.5. Process Design

The key Process Design Deliverables are listed Table 5.4 below.

5.5.1. Highlights

1. Design (PFD and P&IDs) that is based on integration of BFPCL Pipeline Manifold Platforms with SPDC Well Pad Manifolds.
2. Overpressure protection of the Pipelines through relief & shutdown devices.
3. Pipe Flow Hydraulics Analysis using both Pipesim & HYSYS softwares to the benefit of collective assurance and proving of results.
4. Pipeline Network that can handle can handle the GSPA's 340 MMscfd, with capacity of up to 440 MMscfd, and turndown of about 150 MMscfd.
5. Establishment of Flow Assurance for the Pipeline System, through dynamic simulation, using OLGA software.
6. Establishment of maximum slug volume of 7,015 m^3 (44,123 bbl), at 120% of the expected slug volume, which will be the basis of slug catcher sizing.
7. Specification of finger type slug as most suitable for the process.

5.5.2. Preliminary Specifications of the Slug Catcher

Business Case

- Large Volume of Liquid will be encountered, up to 7,015 m^3 (44,124 bbl)
- Initial Separation for the Large Volume of Liquid to ensure efficient operation of inlet separators

Type Selection

- Two-Phase Separation – coarse separation
- Multi-pipe selected as most appropriate following Type Selection Tree (BP Selection Process)
    1. Fluid Type (water gas rather than black oil)
    2. Large Slug Volume of 7,015 m^3 (>> 100 m^3 ref Shell DEP)
    3. Moderate Pressure of 60 – 65 barg, – (14 barg < P < 70 barg, ref BP Type Selection Decision Tree)
    4. Cost Advantage over high pressure vessel design (view smaller wall thickness for pipe compared with vessel for a given pressure)
    5. Greater flexibility in design possibilities to cope with wide ranges of flow situations

5.5.3. Key Process Deliverables

Table 5,4: Key Process Design Deliverables

S/N	Description	References	
1	PFD / P&IDs	PFDs BBOU, ELEP, & AKAB Manifold Platfo	P&IDs - BBOU, ELEP, & AKAB
2	Heat & Material Balance	Heat and Material Balance.xlsx	
2	Pipe Flow Hydraulics Analysis Report - Pipesim	PIPESIM Hydraulics Analysis and Report (!	BFPCL PipeFlow Hydrauilics Parameter
2	Pipe Flow Hydraulics Analysis PFD – BBOU - HYSYS	Base Case_BBOU-ELEP 100	Base Case_BBOU-Elepa 80
2	Pipe Flow Hydraulics Analysis PFD – ELEP - HYSYS	Base Case_ELEP-AKAB 100	Base Case_ELEP-AKAB 80 b
3	Transient Simulation and Flow Assurance Report	Transient Simulation & Flow Assurance Rep	
3	Pipeline Blowdown & Depressurisation	Pipeline Blowdown and Depressurisation	Process Depressurization Stuc

6. Materials Specifications & Corrosion Management

6.1. Specific Key Drivers

The following are specific key drivers to the Materials Specification & Corrosion Management:

1 Selection of suitable materials that will withstand the fluid flow conditions & fluid properties, and the external environment, in respect of material deterioration

2 Specification of corrosion allowance, and corrosion mitigation system that will sustain system's integrity over the life cycle

6.2. Materials Selection Philosophy

The following are factors that have been considered in the processes and procedures for the optimal selection of materials (linepipes, piping, equipment, isolation joints / insulation flanges) for this FEED phase1 project:

- Design Life
- Properties of Internal Fluid
- Surrounding Environment
- Resistance Level to Corrosion Effect
- Inspection & Corrosion Monitoring Possibilities
- Operating & Maintenance Philosophy, and Degree of System Redundancy
- Mechanical Properties
- Wall Thickness
- Weldability
- Temperature Resistance
- Weight Requirement
- Fatigue and Fracture Resistance
- System Availability Requirement
- Fabrication and Installation Methods and Procedures
- Compatibility with Adjoining Materials
- Market Availability of Materials
- Local Experience (both with the materials and the corrosion management system that will apply)
- Economics

The purpose is to establish procedures, specifications and approaches to contend with in the systematic choice of materials and their sub-components for this phase 1 FEED in accordance with international guidelines, codes & standards.

6.3. Pipe Material Types and Selection Processes

Pipe material selection of various components is dependent on the type of materials it transports. The relevant design codes and standards associated with this wet gas transportation are ASME-B31.1, B31.3, B31.8, ASTM-A53, A106, A234, A333, A671/2, API-5L and AWS.

Frequently, the groups used for line pipe and piping materials are:

- High Carbon steel – over 0.6% C in steel
- Medium Carbon steel - between 0.3% to 0.6% C in steel whilst
- Mild Steel is less than 0.3%. C.

For caution, carbon content above 0.35% makes the steel to become brittle and more than 0.43% is a challenge to welding. (Ref. carbon equivalent formula).

The line pipe and piping are graded according to strength (ASTM-A53, ASTM-A106).

- Grade C – Highest Strength
- Grade B – Less Strength
- Grade A – A106 is preferred at high temperature.

The cheap cost price of CS is what makes it useful for engineering applications, however the properties of API specifications 5L or ISO 3183 are better for line pipe production. Again, Carbon Manganese low alloy steel are now more favoured for line pipe materials production because of improved cost.

The processes of selection are now given in flowchart in Fig 4.2 below. The manganese including Nb, V and Ti further improve the precipitation, and strengthen properties by micro alloying of the CS. Hence, this composition is more preferred. Again, it is noted to have more improved quality for corrosion resistance (NACE MR0175 / ISO 15156-1). Some CO_2 corrosion improvement are also noted in comparison. For the purpose of this specification a 25% reduction may be accommodated on Hydrocor calculation at below 60 ^{0}C of operational temperatures (Ref, Analysis of the corrosion scales formed on API 5L X70 and X80 steel pipe in the presence of CO_2)

6.4. Corrosion Analysis

The corrosion analysis process of fig 6.1 shown below has been followed, reflecting inputs from the process and the relevant management of the corrosion actions.

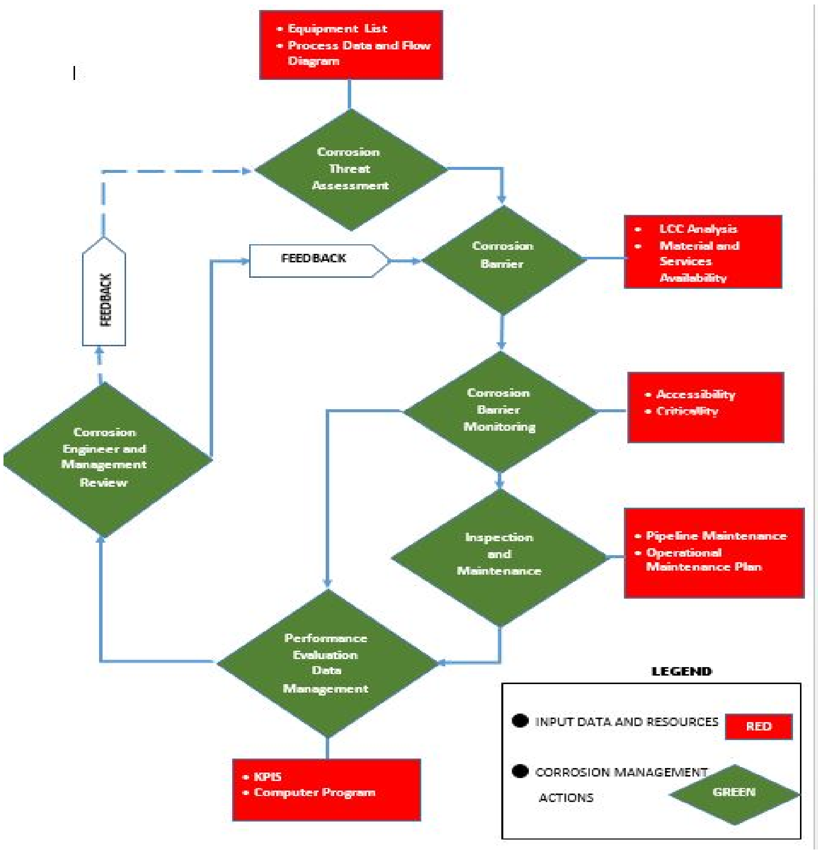


Fig 6.1. Process Flow of Corrosion Analysis, Corrosion Protection and Management Philosophy

This philosophy is in accordance with the NACE provision and approach as shown in the flow process in fig 6.1 above.

This sets out in the following order:

- Identifying the Threats
  - Internal Threats
  - External Threats
- Prediction and Assessment of the Corrosion Damage
- Corrosion Control
- Management of Corrosion and Control Strategies.
- Corrosion Economics

6.4.1. Fluid Composition / Internal Corrosion

The impurities compositions of the wet gas from reservoirs are given in table 6.1 below. The fluid products indicate susceptibility of CO_2, or sweet corrosion. This corrosion threat to MS, was rated for corrosion using initially the De – Waard- Milliams formula and further compared with extrapolation using Nomogram (Ref. fig 6.2) and they were found to be high.

Table 6.1: Reservoirs / Flowing Conditions & Impurities Compositions (Water Saturated Compositions)

	Impurities Compositions - Mole %				
Component	BBOU	ELEP	NUNR	SIEB	OPUG
N_2	0.07	0.00	0.00	0.00	0.07
CO_2	0.59	2.18	0.77	0.77	0.59
H_2S	0.00	0.00	0.00	0.00	0.00
H_2O	0.34	2.19	0.63	0.95	1.08
Total	1.00	4.36	1.40	2.48	4.11
Reservoir / Flowing Conditions					
Res Pres (barg)	223.8	476.6	500.6	450.6	356.9
Res Temp (°C)	78	123	102	117	113.8
FTHP (barg)	100 - 80	100 - 80	100 - 80	100 - 80	100 - 80
FTHT (°C)	62 - 58	63 - 60	63 - 60	63 - 60	62 - 58
pCO_2 (barg)	0.59 – 0.47	2.18 – 1.74	0.77 – 0.62	0.77 – 0.62	0.59 – 0.47

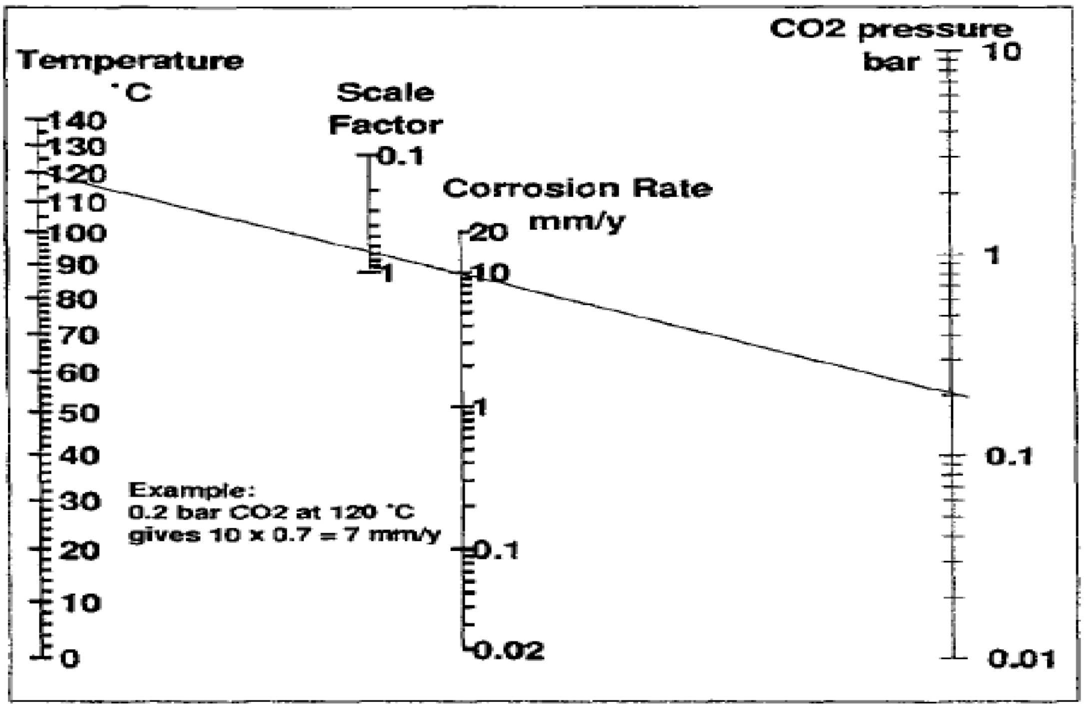


Fig 6.2: DeWaard-Milliams CO_2 Corrosion Nomogram

Again, Hydrocor software assessments confirmed high rates of corrosion (Fig 6.3 and Fig 6.4) in bare MS pipeline carried out for this phase1 FEED. The unmitigated corrosion rates are 1.85 mm/year for the 24”

BBOU – ELEP Line and 3.93 mm/year for the 36" ELEP – AKAB Line. These rates shall result in CA of 55.5mm and 117.5mm unmitigated corrosion allowances for the design life of 30 years, respectively. The rates are undesirable and unacceptable. Hence, corrosion mitigation using inhibition was assessed in accordance with the philosophy in chapter 4 of this document.

The resultant outcome along with the materials' resistance to CO_2 corrosion of 25% of the selected pipeline materials (API specification 5L grade x70 low C-Mn alloy steel) is 3mm and 6mm mitigated CA for the 24" BBOU – ELEP Line and ELEP – AKABEL Line respectively. The reduction in the corrosion levels were quite high, thus justifying the use of the inhibitor. Thus, the use of inhibitor allows us to accommodate the use of the API Grade 5L X70 specification for the pipeline choice.

Consequently, the targeted reduction in CA can be achieved using the recommended Champion X's inhibitor {CORR11447A (EC1458A)} – Champion X is a reputable American chemical company with representation in Nigeria). They selected an appropriate inhibitor that will achieve a target of 92.5% corrosion reduction based on Proforma requisition from Epic Atlantic, following NACE SP 21469 Selection & Management Process for Corrosion Inhibitors.

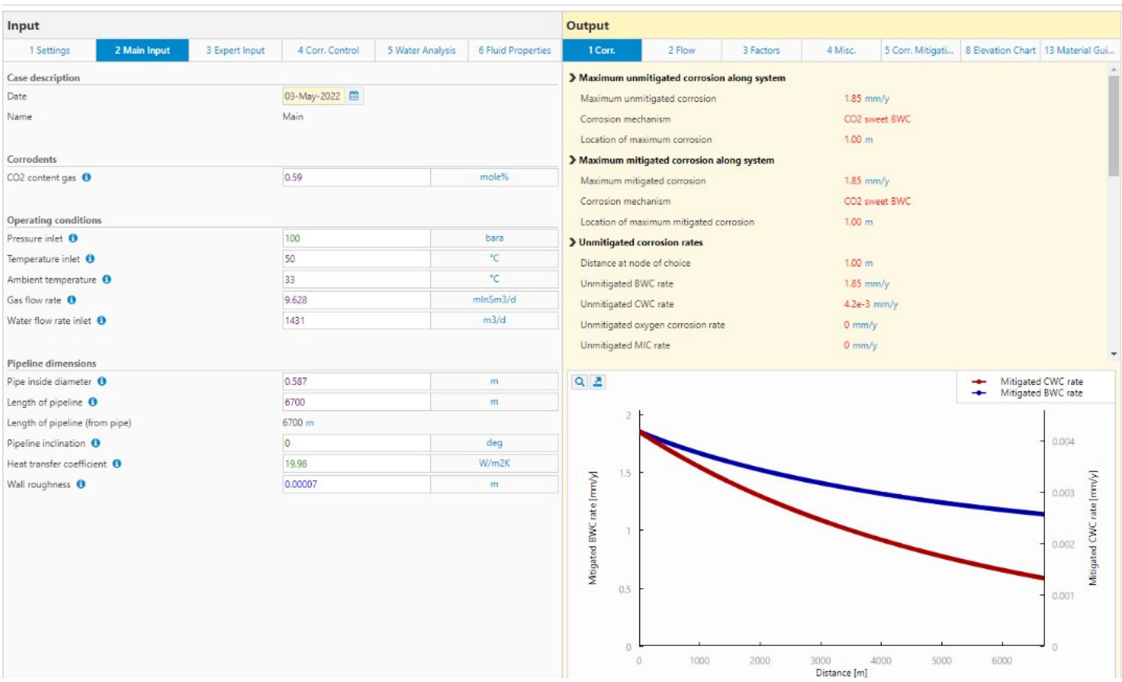


Fig 6.3 Hydrocor Report for 24" BBOU to ELEP Gas Pipeline

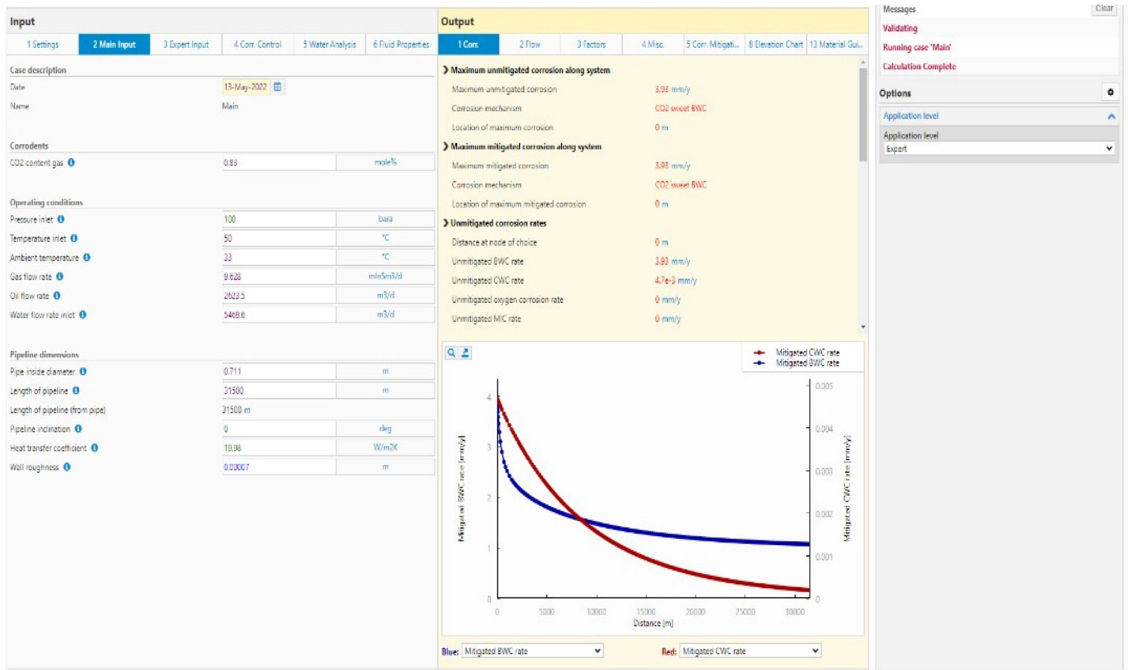


Fig 6.4 Hydrocor Report for 36" ELEP to AKABEL Gas Transmission Line

Environment / External Corrosion

The soil resistivity survey data carried out by the Brone Positioning and Survey Limited, showed that most of the area have very low resistivity reflecting extreme corrosion rates that could be 0.385/0.500 mm/yr. Hence, there is need to arrest the corrosion to the external surface of the pipeline at all times including during construction. Consequently, should there be no CP during the construction period of say three years, there is the possibility that the line will corrode up to 1.5mm and this is undesirable. Conclusively, there is need for temporary CP on the pipelines during the construction period.

The CP can be achieved through two main means:

Sacrificial Anode (S.A) Cathodic Protection System

Impressed Current Cathodic Protection (I.C.C.P) System.

Each of these systems will require other sub level selections of CP items and equipment.

Overall, the intent is to build up the CP type from the choices made from items and equipment to allow for proper CP design.

6.5. Line Pipe Materials Specification

The specified steel grade is API 5L grade X70 PSL 2 – low C-Mn alloy steel, more so for the critical nature of service of the pipeline.

The line pipe Wall Thickness (WT) - is calculated by the addition of the initial WT obtained, and the mitigated CA making the wall thickness of the line pipe process to be finalised.

The company (Champion X) inhibitor code CORR11447A (Old code – EC1458A) was selected.

The material selected for the line pipes is Low Carbon-Manganese alloy steel with Nb, V and Ti in the composition.

This specification shall be in accordance with API 5L & ISO 3138 and the sub components pipe and piping classes shall be in accordance with ASTM, ASME and relevant Shell DEPs on the subject. This piping class and classifications will make it possible to avoid the unnecessary efforts during engineering fabrication, procurement shipping, transportation storing of line pipe, the requirements of ancillary requirements fittings and other equipment shall be linked with appropriate piping class.

6.5.1. The Chemical Composition

The chemical composition of the line pipe API 5L Gr x 70 Sch. 160 is specified according to ASTM A234 S3 applied with the following restrictions in table 6.2 below:

Table 6.2: The Chemical Composition

Specification: API 5L PSL 1/2										
Steel Grade: Grade X70 (L485)										
Chemical Composition (%)							Mechanics Capability			
							Pipe body			Weld seam
C max.	Mn max.	P max.	S max.	V max.	Nb max.	Ti max.	Yield Strength min. (Mpa)	Tensile Strength min. (Mpa)	Elongation (%)	Tensile Strength min. (Mpa)
0.26[e]	1.65[e]	0.030	0.030	F	f	F	485	570	c	570

a. Cu < = 0.50% Ni; < 0.50%; Cr <0.50%; and Mo < 0.15%

b. For each reduction of 0.01% below the specified max. Concentration for carbon, and increase of 0.05% above the specified max concentration for Mn is permissible, up to a max. of 1.65% for grades > B, but < = X52; up to a max. of 1.75% for grades > X52, but < X70; and up to a maximum of 2.00% for X70.

c. Unless otherwise agreed Nb + V < 0.06%

d. Nb + V + TI < 0.15%

e. Unless otherwise agreed.

f. Unless otherwise agreed, NB + V = Ti < 0.15%

g. No deliberate addition of B is permitted and the residual B < 0.001%

6.5.2.

6.5.3. Information and Data for the Pipelines

Information on requirements of the selected materials from the design are tabulated in table 6.3 below:

Table 6.3: Information & Data for the Pipelines

Description	Data	Units
Design Life Time	30	Years
Codes	ASME 31.8, ISO 3183, DEP 31.38,01.84 Gen (Materials Selection Index)	
Service/Duty	Transportation of Wet Gas	
Design Factor	0.6	
Design Temperature	45 to 50	°C
Inlet Pressure	100	barg
Pressure Class	ANSI 900# (Class 900) API 3000	
Test Pressure	1.25 x Design Pressure	psi
Maximum Allowable Operating Pressure	N/A but less than design pressure	
Spacing of Existing Line	No existing line	
Pipe Length (Survey Length),	6.7 / 31.5	km (BBOU-ELEP / ELEP-AKAB)
Pipe DN (Hydraulic Studies)	24 / 36	inch
Wall Thickness from Physical Parameters	15.88 /23.83	mm
Product Fluid	Wet Gas	
Product Impurity/ Water Analysis	No - H_2S, Cl, HCO_3	mole%
Gas Flow Rate (24" / 36")	(340 / 340)	MMscfd
Water Flow Rate (24"/ 36")	(9/16.5)	Mbwpd
Corodant / Corrosion Threats	CO_2 - Sweet Corrosion	
Corrosion Rate (Unmitigated) Hydrocor	1.85 / 3.93	mm/yr
Unmitigated CA	55.5 / 117.9	mm
Mitigating with Inhibitor CA/(Considering the ability of over 25% Corrosion resistance on selected materials)	4.2 (3.) / 8.8(6)	mm
Material Selected 24"/36"	API 5L Gr x 70 Sch. 160	Low C-Mn alloy
Manufacturing Tolerance	+10.5 to – 3.5	%
The length of Line-Pipe joint.	40 (12)	Ft (m)
Number of Line-Pipe to be manufactured (Additional lines covering post manufacturing defects and sparing are not included.	24" BBOU to ELEP line = 560 (Plus 10%= 616) 36" ELEP to AKABEL =2,625 Plus 10% = 2,888)	
Longitudinal welding	to be advised by manufacturer	
Pipe End	BE	

6.5.4. Procurement / Manufacturing Specification

This specification covers the procedures that shall be contended with in the process of manufacturing the line pipes. The following are standards and codes that are needed for the manufacturing:

Table 6.4: Manufacturing Information for Requisitions

	S/N	Items	Description
General	1	Service	Wet Gas Transportation
	2	Vendor / Local Agent	VTA
	3	Manufacturer	VTA
	4	Project Name	Gas Processing and Methanol Complex
	5	Location	BUBOUWE – ELEPA - AKABEL, Bayelsa State
Design Information	6	Design Life	30years
	7	Design Factor	0.6
	8	Design Pressure (barg) Inlet	80 to 100
	9	Design Pressure at Gas Plant (barg)	60 to 65
	10	Pipeline Length (km)	6.7/31.5 (totalling 38.2 km)
	11	Gas Flow Rate (MMscfd)	340
Materials	12	Line pipe Specs (MS)	24"/36" API 5L Gr x 70 Sch 160 /
	13	Materials Standards and Codes	ASTM A53 Type F or Type E/
	14	Line pipe Length (ft/m)	40/12
Manufacturing	15	Process of Manufacturing	VTA (Basic oxygen steel process, electric arc smelting
	16	Longitudinal welding	VTA {ERW (HFI), SWA Helical/Spiral weld}
	17	Mill Tolerance (WT)/Ovality/Length	(+10.5 to -3.5%) / ±1%/ 12m
	18	Marking	The pipe shall be naked to ensure full traceability to melt and heat treatment lot.
	19	Certification	Quality system certify ISO 9001. Inspection witnessing
	20	Tensile testing / hardness / toughness	On cut out specimen
	21	NDT	ES
	22	Hydrostatic Test	Yes
	23	Provision of end cap	Yes

6.6. Piping Materials Specification

All piping materials along with their sub-components are specified in accordance with their piping class and classification according to the following international standards and codes: NACE, DEY, ISO, EEMUA, DNV, ASTM, ASME and DEPs, PD5500, EN13480, BS4515, EN1011,

API510, API620, ISO 15614-5 2004

As for other cases the links from input information from the process studies, the P&ID, identifying numbering methodology, sizing and then the initial thicknesses calculation by mechanical engineer shall be used for the required data.

Then, the corrosion/materials engineer reviews the impact of the environment internally and externally and follows the process contained in the materials selection philosophy to come up with a choice. The choice is complemented with piping classes and other classifications to reduce the unnecessary effort of going through the whole process except on special cases of materials consultancy basis, then the whole process is rounded up. For general usage these standards, - Shell DEPs: 31.38.01.12-Gen – Manual Piping Classes – Refining and Chemicals, 31.38.01.10-Gen – Manual Piping Classes – Basis of Design; ASME pressure rating, etc, covers the specification requirements.

6.6.1. Piping Classes

Piping classes shall be applied in accordance with Shell DEP 31.38.01.11-Gen – Manual – Piping – General Requirements. The identification methods to be used, are given in three parts below:

Part 1: In the ASME rating classes: (1) Class 150, (3) 300, (6) 600, (9) 900, (25) 2500.

Part 2: Very significant to this standard is the number allocated to the materials selected for the project which is given as: 1 – (CS), 2 – (low and intermediate alloy steel), 3 – (SS), 4 – (Aluminum / alloys), 5 – (copper and copper alloys), 6 - (Nickel based alloys), 7 – (Non-metallic materials), 8 – (carbon steel and /or galvanised systems).

Part 3: Indicates a process related to the impact testing and non-impact testing. (ASME B31.3)

6.7. Equipment Materials Specification

The equipment material specification shall provide information and services to the principal, manufacturing and fabricating companies, vendors/agents, construction company's operators, including, - Materials specifications, Standards codes and specification, Interface specification, Test specification, Performance, and Quality specification.

6.8. Corrosion Inhibitor Specification

The purpose is to select an inhibitor to achieve the goal of reducing the corrosion rate by 92.5%

Recommended inhibitor – CORR11447A (EC1458A)

Rate of injection – 41 gal/day for 24" BBOU – 70gal /day for 36" ELEP- AKABEL line

Injection Discharge Practice - Continuous injection

Table 6.6: from Champion X Inhibitor Injection Rates

Corrosion Inhibitor Chemical EC1458A (aka CORR11447A)											
S/N	P/L system	Gas Prod. (MMscfd)	Press. (bar)	Temp. (°C)	Condensate (Mbcpd)	H_2O Prod. (Mbwpd)	CO_2 (mole%)	Corr. Rate (mm/yr)	Diameter (in)	Injection PPM	Injection Rate (gals/day)
1	BBOU to ELEPA	331	100	45-50	16.2	9	0.59	1.85	24	**109**	**41**
2	ELEPA to AKABEL	386	100	45-50	34.4	16.5	0.83	3.93	36	**101**	**70**

6.8.1. Codes and Standards

NACE- 5134-3980 Standard Laboratory Tests and evaluation. This shall also address the residual output of the inhibitor resulting from the injection - DEP 31.01.10.10-Gen: Manual – Chemical Injection Facilities.

6.8.2. Injection Provision

Two chemical injection skids shall be provided, one as back that will ensure 100% availability. The chemical Injection system shall be provided in accordance NACE and other international standards and codes.

The injection system should be designed, constructed, and operated in accordance with NACE standards and codes

The pump shall be rated for this input. The manufacturer's information is given in the Datasheet attached of this document.

6.8.3. Operation Practice

The recommended practice shall be for continuous injection of the inhibitor at the recommended rates.

The inhibitor shall be stored in the tanks provided at the manifold for period agreed and approved by the principal. The storage capacity shall be based on the number days the principal approved to handle the daily rates and the frequencies of replacement of the inhibitor.

Inhibition residual checks and test shall be effected on random sample at the remote end from the pumping locations at Bubouwe and Elepa.

Quality check of product must be certified and guaranteed.

The running of the system shall be done by the operation staff with support from the manufacturer.

Contamination of the inhibitor must be totally prevented and practice for storing the inhibitor shall be done according to approved international standards and codes. SHOC (MSDS) standard for the handling of chemicals shall be strictly followed.

6.9. CP Specification

6.9.1. Sacrificial Anode.

6.9.1.1. Criteria for Cathodic Protection and Overprotection Potentials for Pipelines

Table 6.7 Criteria for Protection and Overprotection Potentials.

S/N	Environment	Reference Electrode		
		$Cu/CuSO_4$(mV)	Ag/AgCl(mV)	Zinc(mV)
1	Soil environment	-950 to -1100	Not used for Soil	Not used for soil
2	Free sea water	-850 to -1000	-800 to -1050	+250 to +0
3	Anaerobic environments	-950 to -1100	-900 to -1050	+150 to +0
4	High strength steels UTS > 700 N/mm^2	-850 to -1000	-800 to -900	+250 to +100

6.9.1.2. Temporary Anode Installation Practice

Table 6.8: Pipeline Data

S/N	PIPELINE	NPS (inch)	LENGTH (m)	AMBIENT TEMP. (°C)	WALL THICKNESS CALCULATED + CA
1.	BBOU - ELEP	24	6,700	35	12.88 + 3 = 15.88
2.	ELEP - AKAB	36	31,500	35	17.83 + 6 = 23.83

6.9.1.3. S.A Material Prequalification Requirements.

The Mg anode material being considered shall be pre-qualified to ensure their performance (resistivity potential as electro chemical capacity) at the operating temperature and soil conditions. Certified test reports from the proposed anode foundry or a certifying agency approved by the principal, or for previously cast anodes of identical chemistry may be submitted in lieu of pre-qualification testing specifically for this project approval of the company in advance of casting the sacrificial anodes.

6.10. Impressed Current Cathodic Protection Type Specification.

This specification and procedures shall satisfy the need for the FEED and installations of the following equipment

- Groundbed
- Transformer – Rectifiers
- Cable Makers
- Junction Boxes
- Test Post

6.10.1. Groundbed -Deepwell

6.10.1.1. General

A good performance of the CP system will be obtained by deep well ground bed type.

This will allow for satisfactory, spread of protective current to the pipelines. The ground bed will comprise Mixed Metal Oxide – (MMO) based anodes Type.

Schematic Sketch below (fig 13.1)

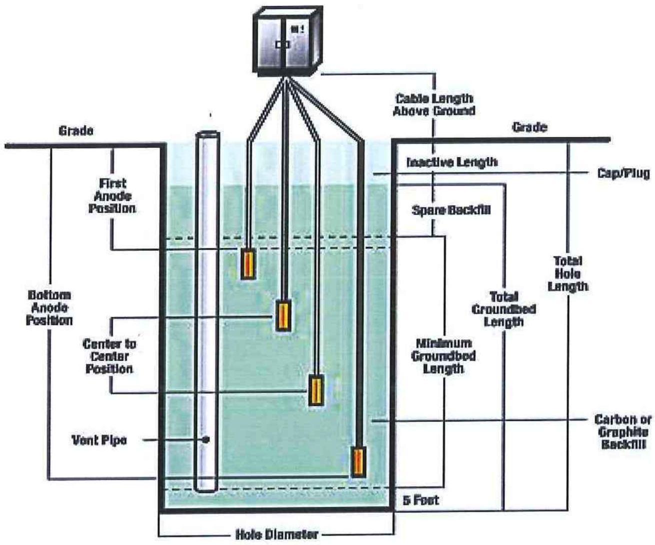


Fig 13.1: Schematic Sketch of Deepwell Groundbed

The Groundbed shall be designed according to the DEP – 30.10.73.31-Gen and NACE.

NB: The design life of the CP is 30 yrs.

The CP contractor shall decide the size of the borehole.

Each anode shall be mounted on a single self – supported single core 6 mm^2 EPR/PV Cable at the manufacturer's workshop.

The cable will be of a type suitable for installations in a borehole filled with carbonaceous backfill. Each anode cable shall be terminated separately at the anode bus bar.

the manifold.

6.11. Key Materials Specification and Corrosion Management Deliverables

Table 6.2: Key Materials Specification & corrosion Management Deliverables

S/N	Description	References	
1	Materials Specification & Corrosion Management	PDF Material Selection & Corrosion Manageme	
2	Pipeline Cathodic Protection Layout	PDF Pipeline Cathodic Protection Layout.pdf	

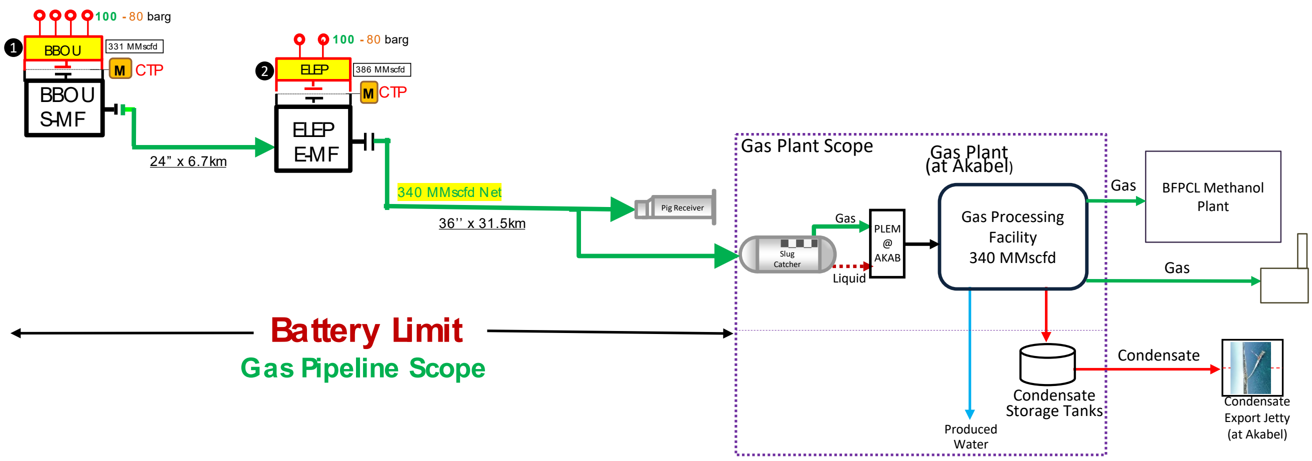
7. Pipeline Engineering

7.1. Specific Key Drivers / Decissions

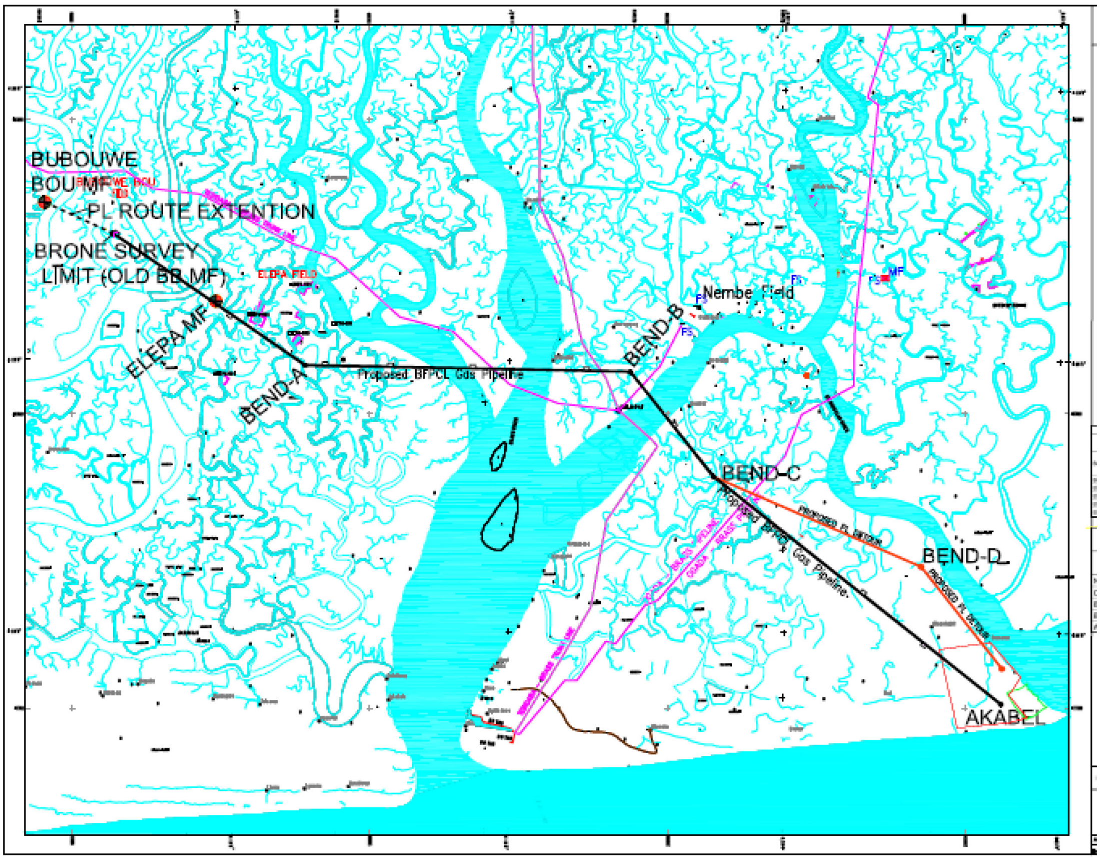
The following are specific key drivers / decisions to the Pipeline Engineering:

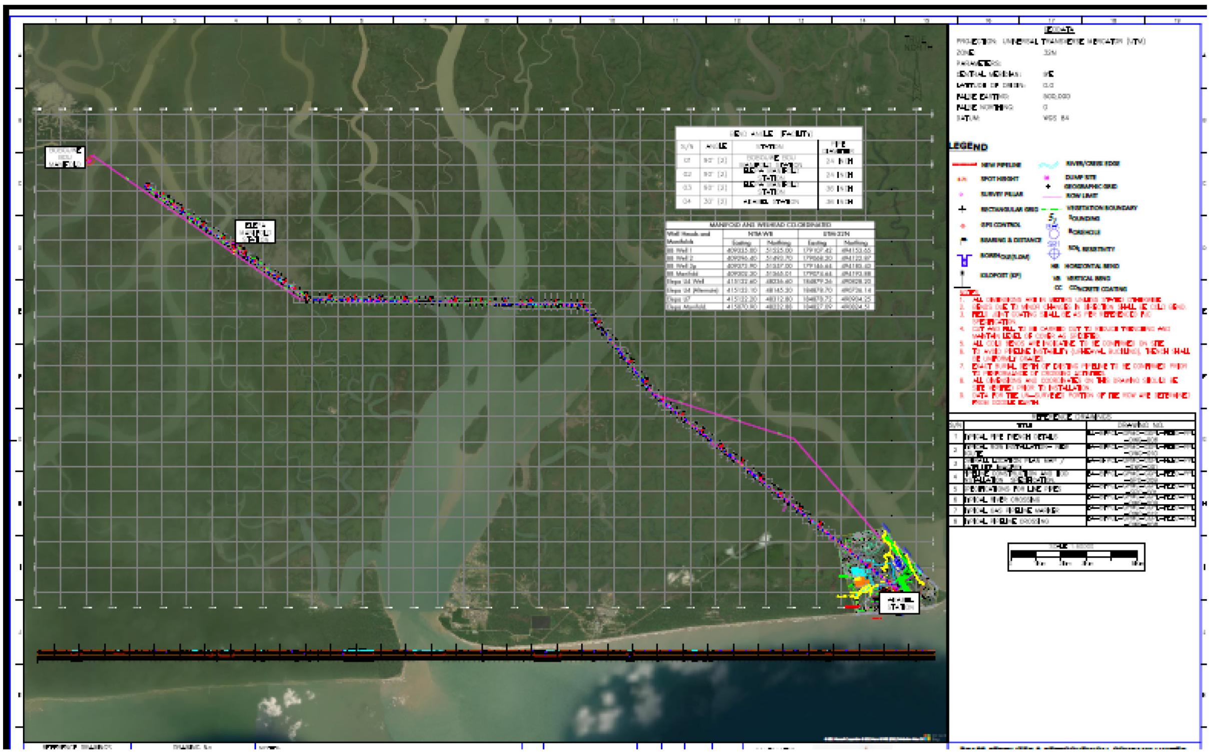
1. Pipeline sizing that will deliver at the required pressure without the need for booster compression
2. Specification of wall thickness of line pipe and piping class capable of withstanding design pressure
3. Selection of suitable materials that will withstand the fluid flow conditions & fluid properties, and the external environment, in respect of material deterioration
4. Specification of corrosion allowance, and corrosion mitigation system that will sustain system's integrity over the life cycle
5. Concrete coating throughout the entire pipeline length for negative buoyancy (swampy terrain) and physical protection
6. HDD at River Crossings
7. Submarine Power & Fibre Optic Cables in concurrent lay with the Pipeline and along the same ROW

7.2. The Pipelines



7.3. Route Survey Map / Overall Location Plan





7.4. Pipeline Crossing Schedule

BUBOUWE MANIFOLD STATION TO ELEPA MANIFOLD STATION					
S/N	START KP (Km)	END KP (Km)	CROSSING	LENGTH OF CROSSING (m)	PROTECTION TYPE
1	0+450	0+780	RIVER	330	HDD
2	1+560	1+600	CREEK	40	CONCRETE COATING
3	2+060	2+120	RIVER	60	CONCRETE COATING
4	3+560	3+600	CREEK	40	CONCRETE COATING
5	4+040	4+080	CREEK	40	CONCRETE COATING
6	4+200	4+480	RIVER	280	HDD
7	5+440	5+520	CREEK	80	CONCRETE COATING
8	5+700	6+080	RIVER	380	HDD
9	6+660	6+700	CREEK	40	CONCRETE COATING

ELEPA MANIFOLD STATION TO AKABEL PIGGING STATION					
S/N	START KP (Km)	END KP (Km)	CROSSING	LENGTH OF CROSSING (m)	PROTECTION TYPE
1	0+520	0+600	CREEK	80	CONCRETE COATING
2	1+100	1+260	CREEK	60	CONCRETE COATING
3	2+100	2+600	CREEK	500	CONCRETE COATING
4	3+000	3+200	CREEK	200	CONCRETE COATING
5	3+340	3+420	CREEK	80	CONCRETE COATING
6	3+780	3+920	RIVER	140	CONCRETE COATING
7	4+240	4+300	CREEK	60	CONCRETE COATING
8	5+020	5+080	CREEK	60	CONCRETE COATING
9	6+260	6+320	CREEK	60	CONCRETE COATING
10	6+360	6+480	CREEK	120	CONCRETE COATING
11	6+920	6+960	CREEK	40	CONCRETE COATING
12	8+300	9+500	RIVER	1200	HDD
13	10+246	10+266	TEBIDADA - BRASS TRUNK PIPELINE	20	
14	11+160	12+260	RIVER	1100	HDD
15	13+780	13+860	CREEK	80	CONCRETE COATING
16	13+879	13+902	SHELL AGIP PIPELINE	20	
17	15+454	15+474	SHELL AGIP PIPELINE	20	
18	15+840	16+900	RIVER	1060	HDD
19	17+560	17+720	CREEK	160	CONCRETE COATING
20	18+160	18+240	RIVER	80	CONCRETE COATING
21	18+960	19+000	CREEK	40	CONCRETE COATING
22	19+500	19+720	RIVER	220	HDD
23	20+560	20+640	RIVER	100	CONCRETE COATING
24	20+870	20+920	OGADA - BRASS PIPELINE	50	
25	21+160	21+280	CREEK	120	CONCRETE COATING
26	21+400	21+680	CREEK	280	CONCRETE COATING
27	22+100	22+700	RIVER	600	HDD
28	23+000	23+220	CREEK	220	CONCRETE COATING
29	24+900	25+140	CREEK	240	CONCRETE COATING
30	25+300	23+880	CREEK	380	CONCRETE COATING
31	26+080	26+140	CLEARED/ PROPOSED NOAC PIPELINE ROW	60	
32	26+820	26+940	CREEK	120	CONCRETE COATING
33	28+440	28+800	RIVER	360	HDD
34	30+400	30+720	CREEK	320	CONCRETE COATING
35	31+040	31+120	CREEK	80	CONCRETE COATING

7.5. Pipeline Technical Specifications

24" x 6.7km BBOU - ELEP

Parameter	Description
Design Code	ASME B31.8
From	BBOU Manifold Station
To	ELEPA Manifold Station
Pipeline Length	6.7km
Design Life	30 years
Process Fluid	Condensate / Saturated Hydrocarbon Gas.
Design Pressure	110 barg
Operating Pressure	100 barg
Max Design Temperature	60°C
Min Design Temp (Pipeline)	12.7°C
Min Design Temperature (Vent Line)	-13.3°C
Line Pipe Material Grade	API 5L–X70(PLS2) SAWL - 100% radiographed.
Installation Temperature	23°C
Specified Minimum Yield Strength	483 Mpa
NPS	24"
Density	7,850 kg/m3
Young's Modulus of elasticity	2.07 x 10^5 MPa
Poisson's ratio	0.3
Coefficient of linear thermal expansion	11.7 x 10-6 m/ m/ °C
Ultimate Tensile Strength	570 MPa
Steel Specific Resistivity	0.018 Ω m/mm2
Wall Thickness	15.88mm
Design Factor	0.6
Corrosion Allowance	3 mm
ANSI Class	900#
Parent Pipe External Coating	3LPE
Concrete Coating Thickness	60mm
Installation Condition	Trenching
Cathodic Protection	Impressed current

<u>36” x 31.5km BBOU - ELEP</u>

PARAMETER	DESCRIPTION
From	ELEPA Manifold Station
To	AKABEL Station
Pipeline Length	31.5km
Design Life	30 years
Process Fluid	Condensate/hydrated Hydrocarbon Gas.
Design Pressure	110 barg
Operating Pressure	100 barg
Max Design Temperature	60°C
Min Design Temp (Pipeline)	12.7°C
Min Design Temperature	-13.3°C
Line Pipe Material Grade	API 5L–X70 (PLS2) SAWL - 100% radiographed.
Installation Temperature	23°C
Specified Minimum Yield Strength	483 Mpa
NPS	36"
Density	7,850 kg/m3
Young’s Modulus of elasticity	2.07 x 10^5 MPa
Poisson’s ratio	0.3
Coefficient of linear thermal expansion	11.7 x 10-6 m/ m/ °C
Ultimate Tensile Strength	570MPa
Steel Specific Resistivity	0.018 Ω m/mm2
Wall Thickness	23.83mm
Design Factor	0.6
Corrosion Allowance	6 mm
ANSI Class	900#
Parent Pipe External Coating	3LPE
Concrete Coating Thickness	85mm
Installation Condition	Trenching
Cathodic Protection	Impressed current

7.6. The Pig Launcher & Receiver

7.6.1. General

- The following description is intended to indicate the general and minimum requirement for the launcher and receiver and does not relieve the supplier of his full responsibility for fabrication performance and safety of the equipment.
- Trap assemblies shall be suitable for launching or receiving pigs for the purpose of inspection, gagging, cleaning and removing liquids from the pipeline which may contain water and/or liquid hydrocarbons and impurities such as sand and scale.
- Except lifting lugs, lifting facilities, such as runway beam, pig trays, may not be required unless demanded by the company.
- The corrosion allowance is as specified in the data sheets.

7.6.2. PIG Trap System Component

Barrel, reducer and spool

- The spool piece shall have the same nominal size as the connecting pipeline, flanged or bevelled end, as specified in requisition.
- Unless otherwise specified, the barrel shall have a sufficient length to accommodate three pipeline cleaning pigs.
- Barrels shall be designed in accordance with ANSI B 31.8
- A thermal relief line shall be provided at locations where shut-in pressure of trapped fluid could exceed the design pressure.
- The reducer should be eccentric for launcher and be concentric for receiver.

7.6.3. Supports

- Permanent supports shall be used to support and restrain the pig traps and these shall be designed to carry the weight of the pig trap system filled with water (or other fluids if their density is greater than that of water) together with the weight of pigs. The support under the barrel should normally be of the sliding/clamp type to compensate for expansion of the unrestrained part of the pipeline. The frictional force should be reduced to minimum by appropriate means.
- The height of the saddle should be such that the bottom of the barrel located 800 mm. above ground level.
- Supports shall be positioned such that the pig trap valves can be removed for maintenance or replacement without removal of barrel.
- Design of supports shall comply with the requirements of ASME Sec.VIII
- Base plate, anchor bolts, etc. required for foundations shall be adequately design and all the applied forces and moments on the saddles shall be specified and provided for the manufacturer. Where necessary design data shall be requested from the manufacturer.
-

7.6.4. Branch Connections

- The trap shall be completed with all required connections.
- Weldolet and Threadolet are only allowed for connections equal or smaller than 2 inches. Connections larger than 2 inches shall be extruded outlets (reducing tees) or sweepolets if approved by company.

7.6.5. End Closure

- The end closure shall conform to the general requirements of ASME VIII division 1 section UG-35(b) (quick actuating closures). End closure door shall meet the requirements for a fail-safe design of the opening mechanism, specifically; the failure of any part of the opening mechanism shall leave the closure closed rather than open.
- The end closure shall be of the quick acting type, lever or hand wheel operated, and hinged vertically.
- The quick acting design should allow opening and closing by one man in approximately one minute, without the use of additional devices.
- The design of the end closure shall be suitable for permanent location in open environment.
- The activation of the seals shall be such that the fluid within the pig trap at any pressure between 1 bar (abs) and the pig trap design pressure be sufficient for this purpose.
- End closures with exposed screw expanders or captive ratchet braces should not be used, because of the high maintenance requirements and the non-fail-safe aspects of some opening mechanism designs.
- The end closure shall have the following safety devices:

a. A pressure locking device to prevent opening of the door when the pig trap is pressurized.

b. A safety bleeder that when released will alert the operator to a possible hazard unless pressure in the pig trap is relieved completely. Opening of the door shall not be possible unless the bleeder is released. Engaging the bleeder shall only be possible when the closure is closed. The bleeder shall be designed such that there is no risk of blockage. In very toxic service, the pressure safety device shall be non-venting.

c. The necessity for interlocking shall be decided by the Client.

7.6.6. Materials

Pipe for major and minor barrels shall be API 5L X70 unless otherwise specified in the specification for pipeline and/or the specification for pipes.

Flanges shall be weldneck type. All flanges and fittings shall be in accordance with the project requirement.

Studs and nuts shall be PTFE coated. The materials shall be ASTM A193/ grade B7 and ASTM A194/A194M grade 2HM

7.6.7. Fabrication

- Welding processes, welding procedures, welder qualifications, weld repairs, welding electrodes; thermal stress relief and heat treatment, etc. shall conform to the relevant project QAQC requirements.

- All main seam welds and nozzle to body welds shall be full penetration
- The inside of the trap shall be free from obstructions which could prevent the free rolling of spheres, or travel of pigs or carriers.
- Wall thickness transitions shall meet the welding configuration requirements as specified in the ANSI B31.8.

7.6.8. Inspection and Testing

Inspection and testing shall be performed before any coating or paint is applied.

All components shall be visually examined in accordance with ASME VIII division 1, part UG93.

Each weld on the trap and pig signaller shall be examined by 100% radiography (RT) and /or 100% magnetic particle examination (MT) or as may be required by the relevant project procedures.

For the end closure, all mating clamp and flange machined surfaces, door hinge, hinge attachments and locking mechanisms shall be subject to magnetic particle inspection. Any defects shall constitute a basis for rejection.

The trap shall be hydrostatically tested to 1.4 times the design pressure. The test pressure shall be held for a period of at least 4 hours. The acceptance criteria are no leakage or loss in pressure.

7.6.9. Surface Preparation and Finish

After acceptance of hydrostatic test, all external surfaces shall be prepared and prime coated in accordance with project blasting and painting procedures. The suppliers' proposed method of surface preparation shall be approved by the company.

All unpainted surfaces, e.g. flange surfaces, shall be properly protected against corrosion with anti-corrosion compound, easily removable by hydrocarbon solvents.

The pig launching and receiving trap including pig loading and unloading devices and also pig signaller shall be cleaned and shall be painted in accordance with project specification.

7.6.10. Pipeline Pig Passage Detector

The signaller shall be intrusive type and mounted on the trap/pipeline via a DN 50 branch connection.

For pipelines which cannot be depressurized for pig signaller maintenance, the signallers shall be complete with a ball valve to isolate the pig signaller from the pipeline and a portable jacking tool for safe lifting of the transfer mechanism.

7.6.11. Nameplates and Labelling

Each pig launching and receiving trap shall be labelled with engraved stainless or non-corrosive alloy nameplates together with non-corrosive fixing materials, showing all data as called for in this specification including but not limited to, the followings:

d. Company name and order number.
e. The year of manufacture.
f. Manufacturer's name or trade mark.
g. Type of materials and size
h. Flange pressure rating.
i. Design and test pressure.
j. Dimensions and physical properties including weight.
k. Tag number.
l. Design temperature.

7.7. Key Pipeline Engineering Deliverables

Table 7.1: Key Pipeline Engineering Deliverables

S/N	Description	References	
1	Pipeline Alignment Sheets	Pipeline Alignment Sheets 01 to 38.zip	
2	MTO – Long Lead Items	MTO for Long Lead Item IFR.xlsx	
3	Critical Line List - BBOU	Ctitcal Line List-BBOU - FEED Overvie	
4	Critical Line List - ELEP	Ctitcal Line List-ELEPA - FEED Overvie	

8. Mechanical & Piping

8.1. Specific Key Drivers / Decissions

The following are specific key drivers / decisions to the Pipeline Engineering:

1. Design integration between SPDC and BFPCL in respect of Wellhead Manifold Platforms.
2. Placement of the facilities at a height that is reasonably above flood level that will ever be experienced in the area, – drawing lessons from the 2012 rare flood experience
3. High Security Fencing (e.g. ClearVu, Citadel, Herass)

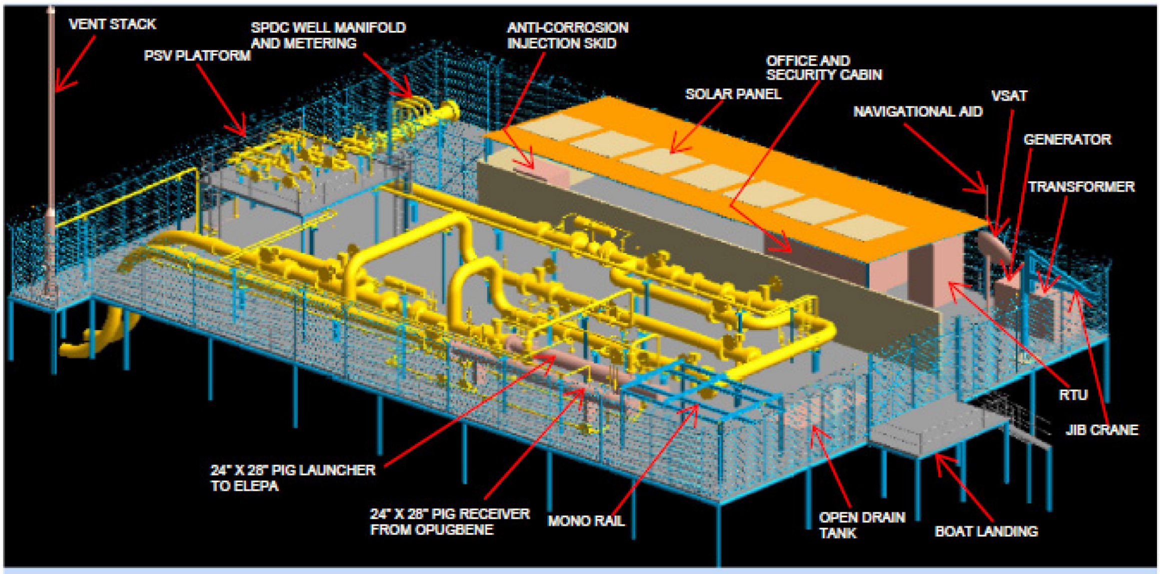
8.2. Scope

Scope	
Pipeline, - complete with: 1. Temporary cathodic protection system (sacrificial anodes) during construction 2. Fibre Optic Cable with pipeline leak and intrusion detection 3. Power Cable 4. Impressed Current Cathodic Protection System 5. Pig Launcher / Receiver	**Akabel Manifold, - complete with:** 1. Pig Receiver 2. Slug Catcher – GPP Scope 3. Earthing System and lightning protection
Boubuwe (BBOU) Manifold, - complete with: 1. Field Manifold Headers – SPDC Wells & BFPCL Bulk 2. Incoming OPUG-BBOU Bulkline / Pig Receiver & Outgoing BBOU-ELEP Bulkline / Pig Launcher 3. RTU - Control Room / Field Auxilliary Room (FAR) 4. Cable Power from Akabel, with Solar as back-up 5. Electrical Transformer & Electrical Room 6. Chemical (Corrosion Inhibitor) Injection System 7. Closed Drain Vessel & Open Drain Tank 8. Weather Shelter / Guard Hut 9. Gantry & Accessories 10. High Security Fencing (e.g. Citadel, Herass) 11. Navigation Aid (lighting) 12. Earthing System and lightning protection 13. Cold Vent (for small inventory trap)	**Elepa (ELEP) Manifold, - complete with**: 1. Field Manifold Headers – SPDC Wells & BFPCL Bulk 2. Incoming NUNR-ELEP Bulkline / Pig Receiver & Outgoing ELEP-AKAB Bulkline / Pig Launcher 3. RTU - Control Room / Field Auxilliary Room (FAR) 4. Cable Power from Akabel 5. Electrical Transformer & Electrical Room 6. Chemical (Corrosion Inhibitor) Injection System 7. Closed Drain Vessel & Open Drain Tank 8. Weather Shelter / Guard hut 9. Gantry & Accessories 10. High Security Fencing (e.g. Citadel, Herass) 11. Navigational Aid (lighting) 12. Earthing system and lightning protection 13. Cold Vent (for small inventory trap)

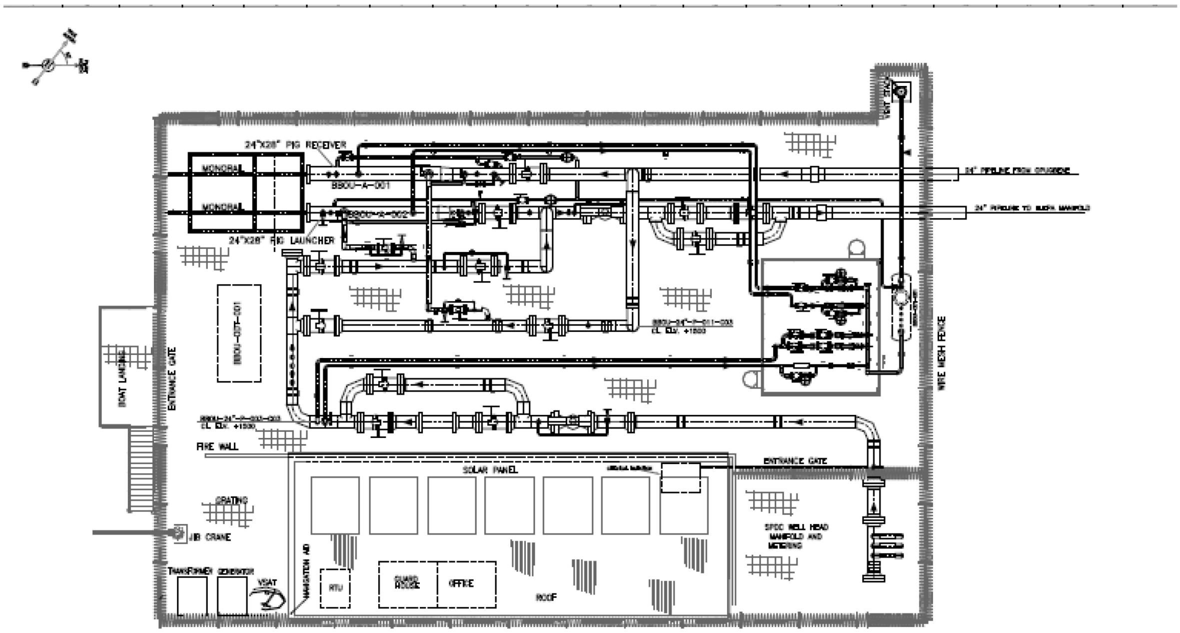
8.3. The Manifolds

BBOU Manifold

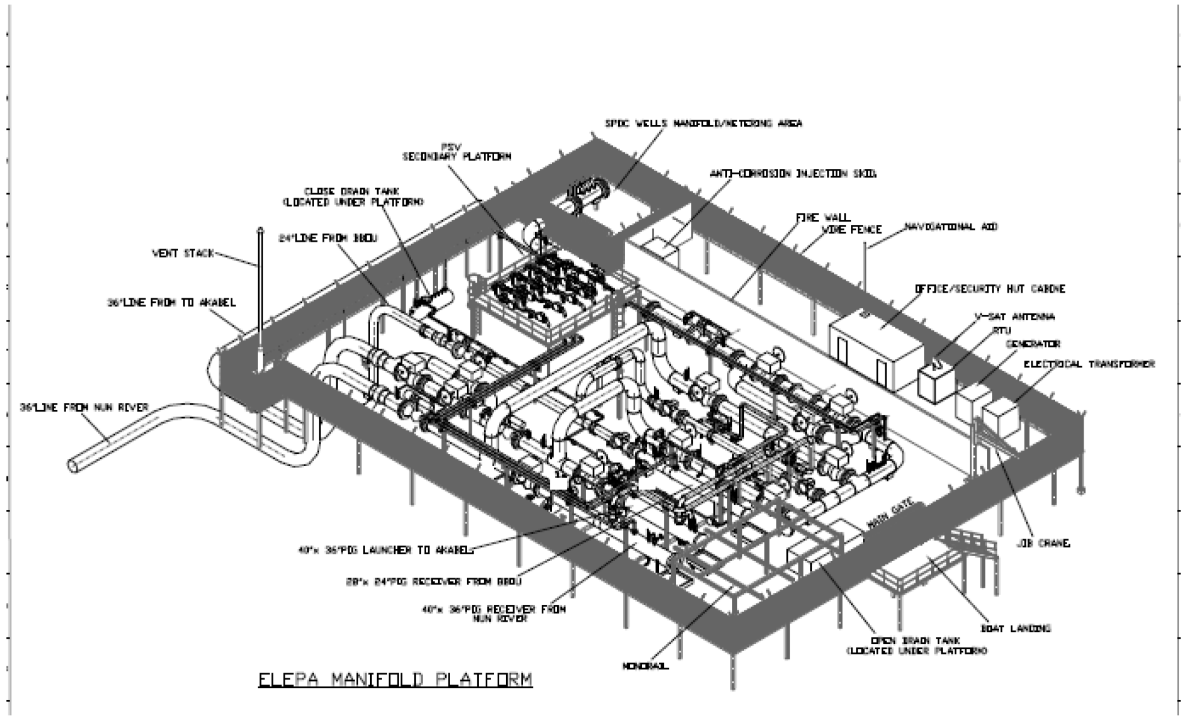
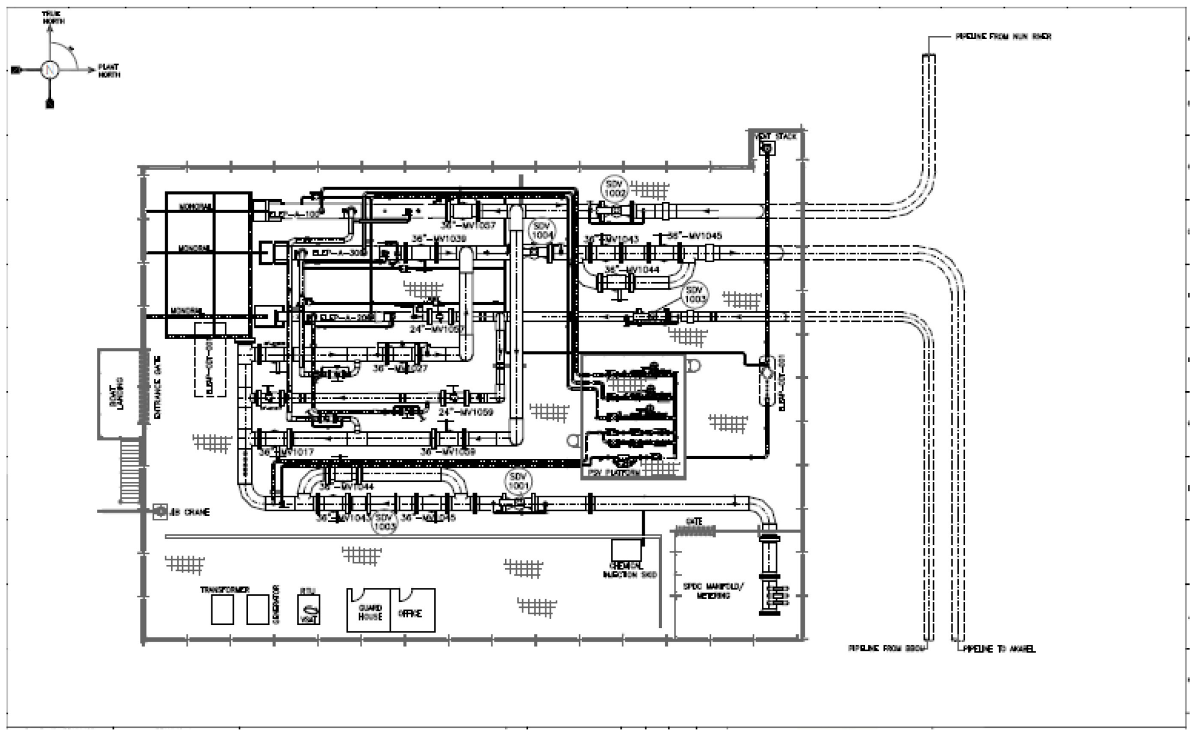
3D Model



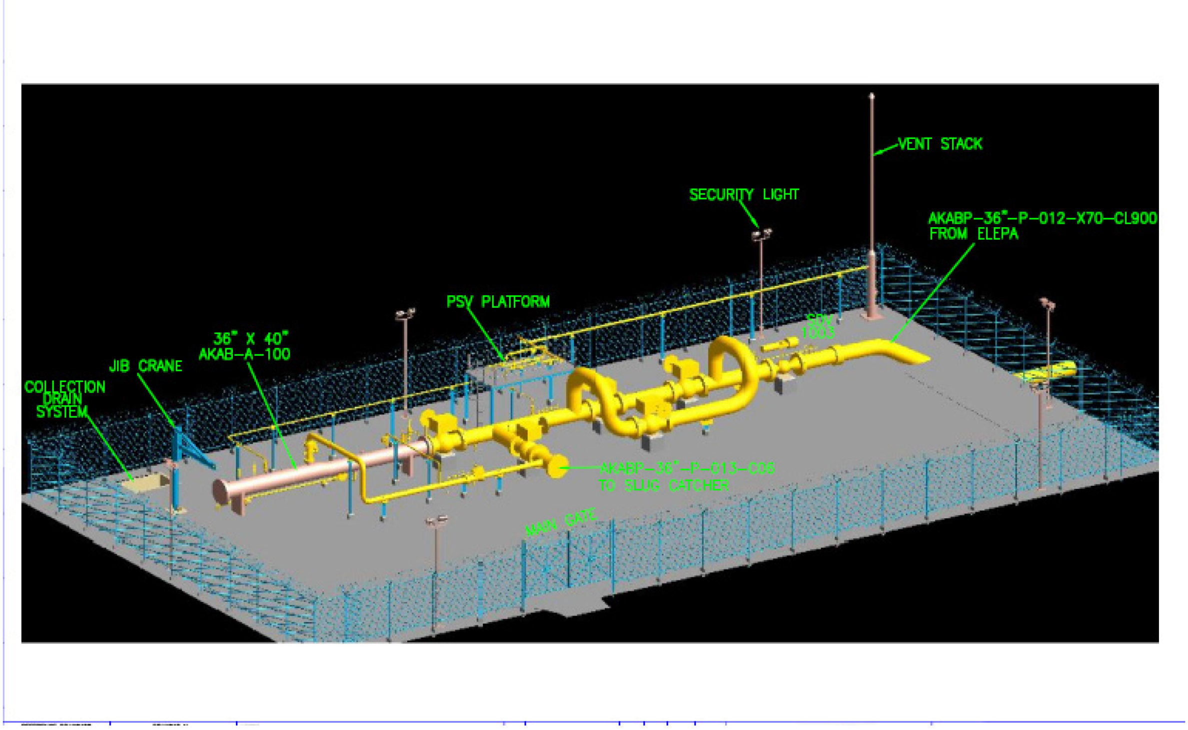
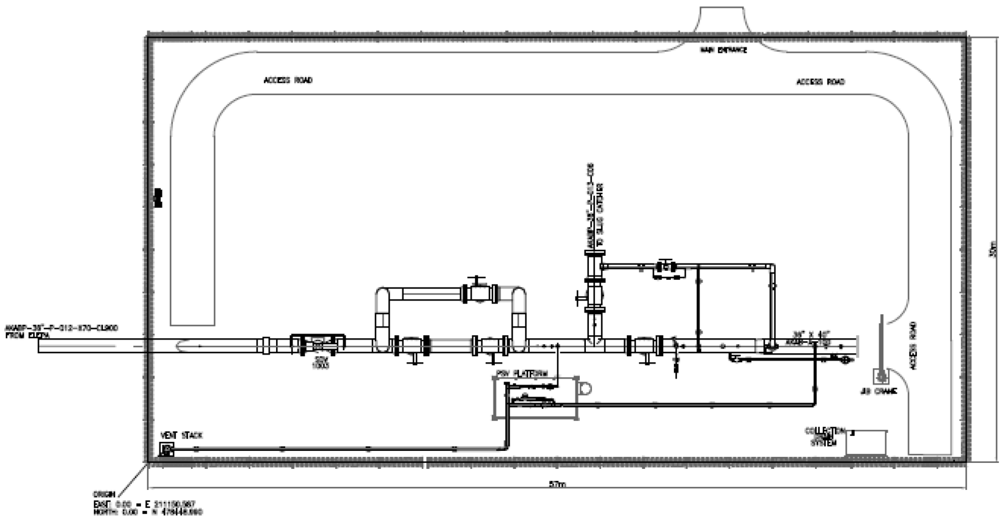
Piping General Arrangement Drawing



ELEP Manifold

3D Model

Piping General Arrangement Drawing


AKAB Manifold

3D Model

Piping General Arrangement Drawing


8.4.

8.5. Key Mechanical & Piping Deliverables

Table 8.1: Key Mechanical & Piping Deliverables

S/N	Description	References	
1	Pipeline Stress Analysis - BBOU	Piping Stress Analysis - BBOU.zip	
2	Pipeline Stress Analysis - ELEP	STRESS ANALYSIS REPORT-ELEPA.pdf	
3	Pipeline Stress Analysis - AKAB	Piping Stress Analysis & Report-AK	
4	Specifications for Pig Trap End Closure	Specifications for Pig Trap End Closure	

9. Structural Design / Civil and Foundation Engineering

9.1. Specific Key Drivers / Decissions

The following are specific key drivers / decisions:

1. Power supply to the manifolds from Akabel power plant via submarine power cable
2. Solar power system at BBOU as back-up to establish viability of the system for subsequent application in the remaining locations

Table 9.1: Structural / Civil & Foundation

S/N	Description	References	
1	Pile Foundation Layout Drawings / Pedestal Layout Drawings	Pile Foundation Layout Drawings.pdf	
2	Structural Layout Drawings	Structural Layout Drawings.pdf	
3	Civil & Structural Calculations	Civil & Structural Calculations.pdf	
4	Civil Works Specifications	Civil Works Specifications.pdf	
5	Structural Works Specifications	Structural Works Specifications.pdf	
6	Sewer System Planimetric Flow diagram	Sewer System Planimetric Flow Diagı	

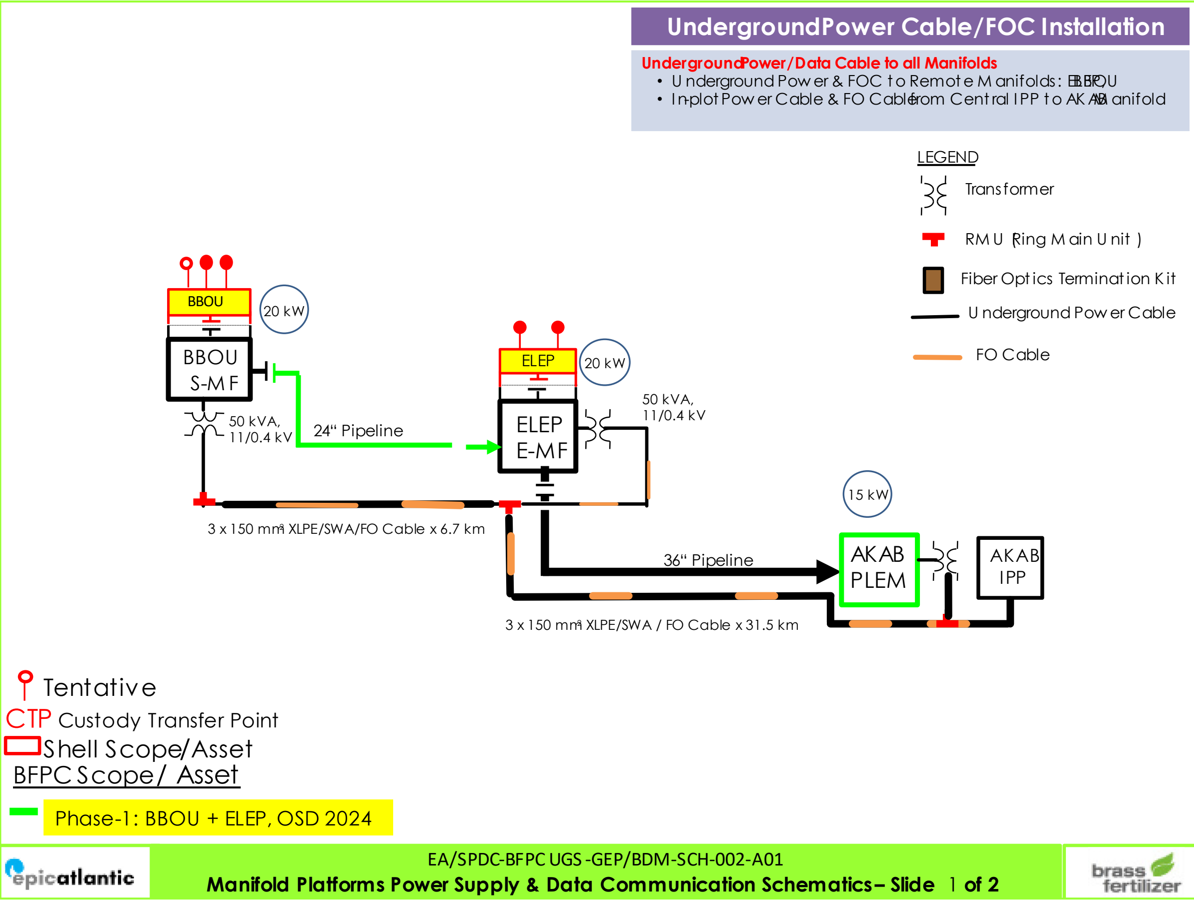
10. Electrical Engineering

10.1. Specific Key Drivers / Decissions

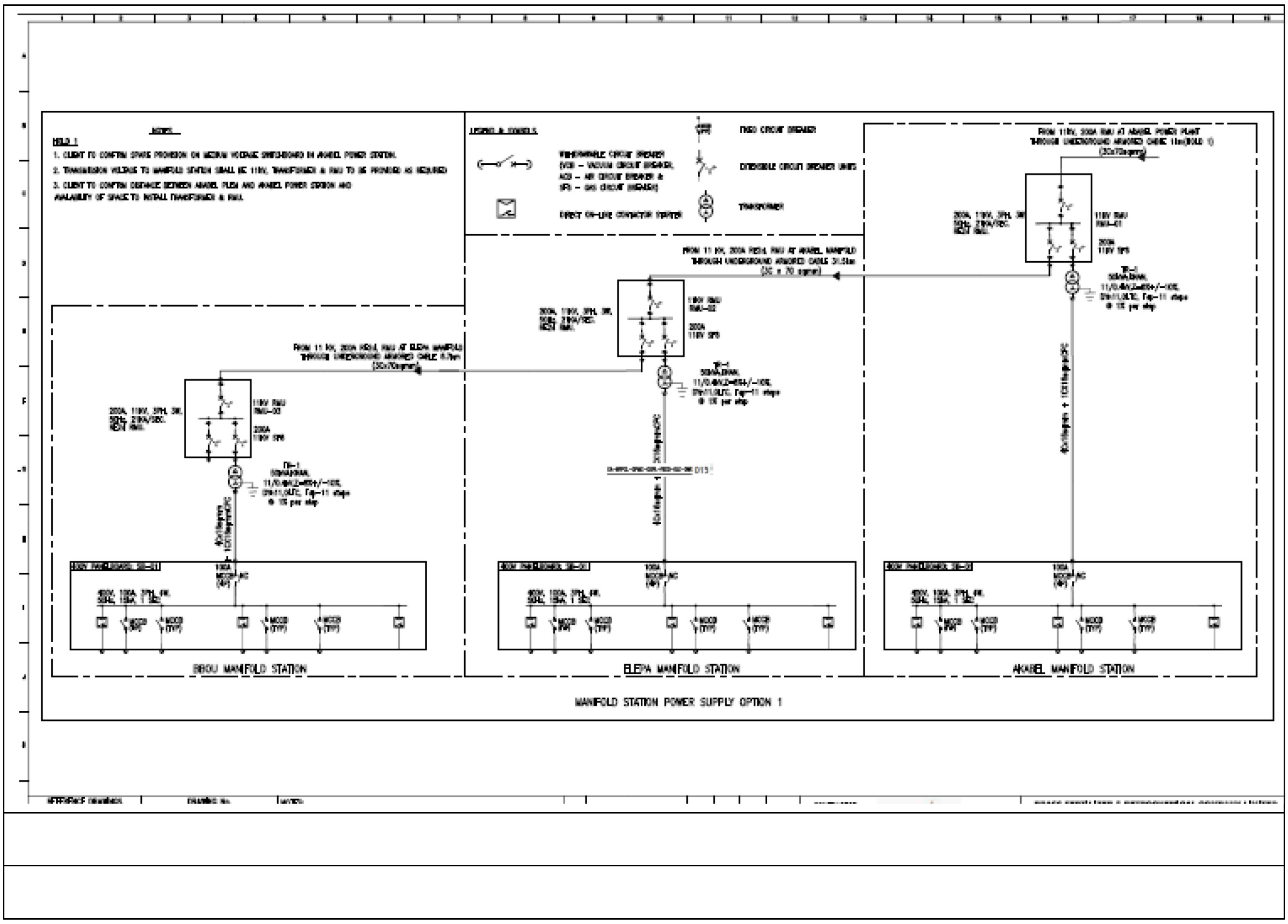
The following are specific key drivers / decisions to the Electrical Engineering:

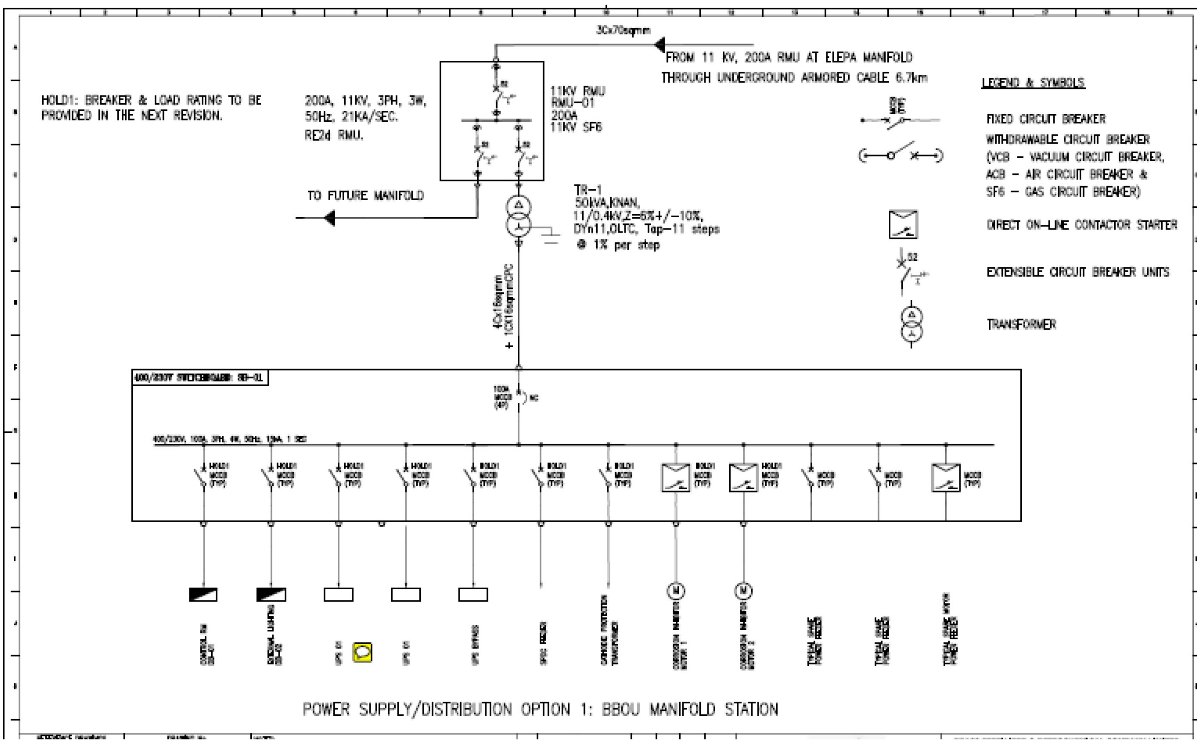
1. Power supply to the manifolds from Akabel power plant via submarine power cable
2. Solar power system at BBOU as back-up to establish viability of the system for subsequent application in the remaining locations

10.2. The Power Supply Schematics

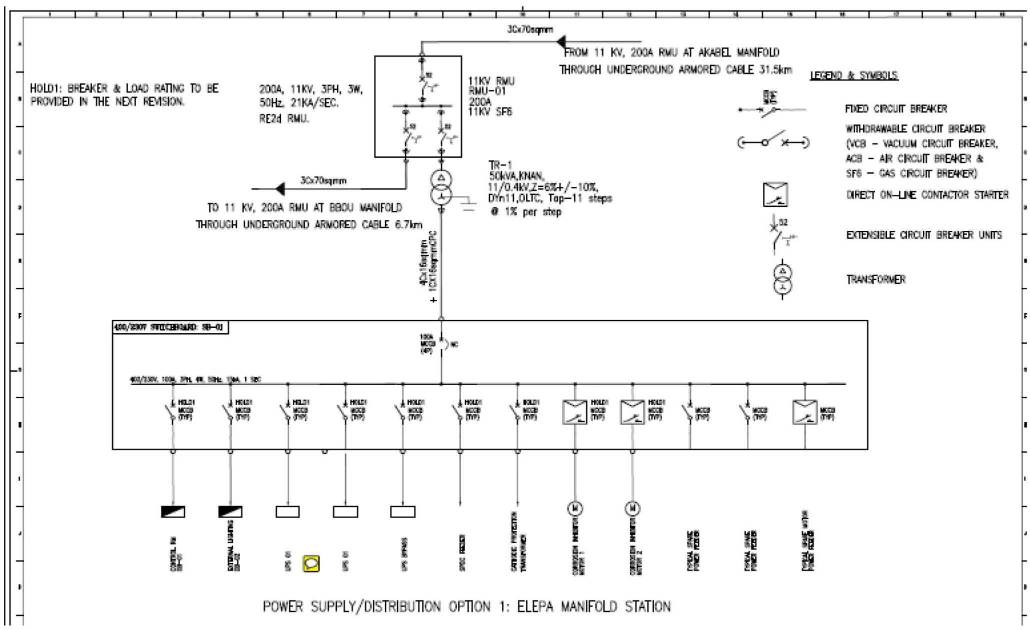


10.3. The Power System Layout

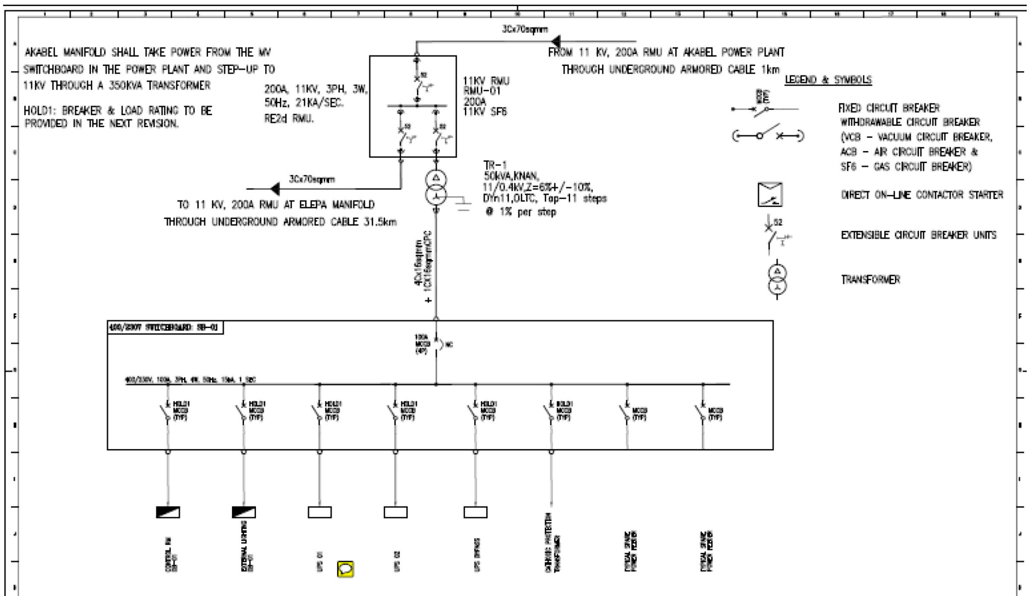




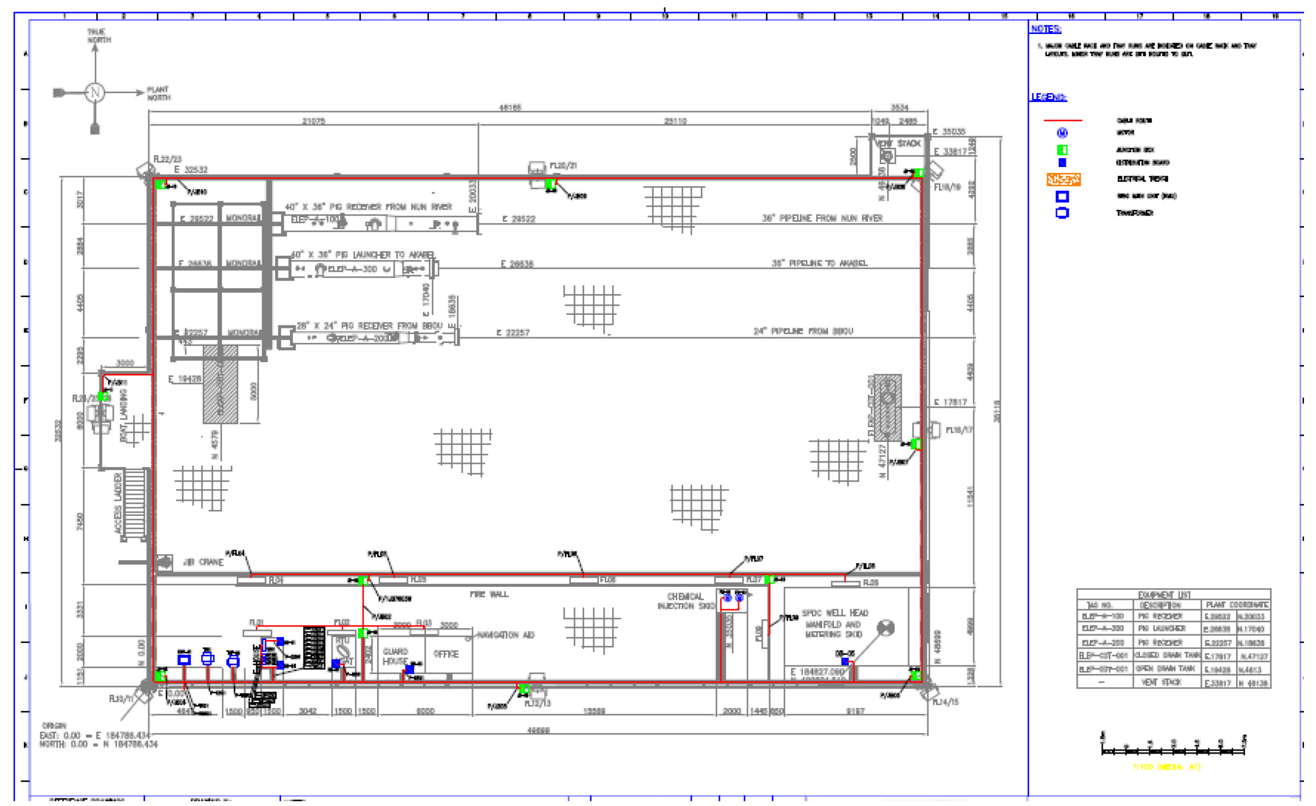
POWER SUPPLY/DISTRIBUTION OPTION 1: BBOU MANIFOLD STATION



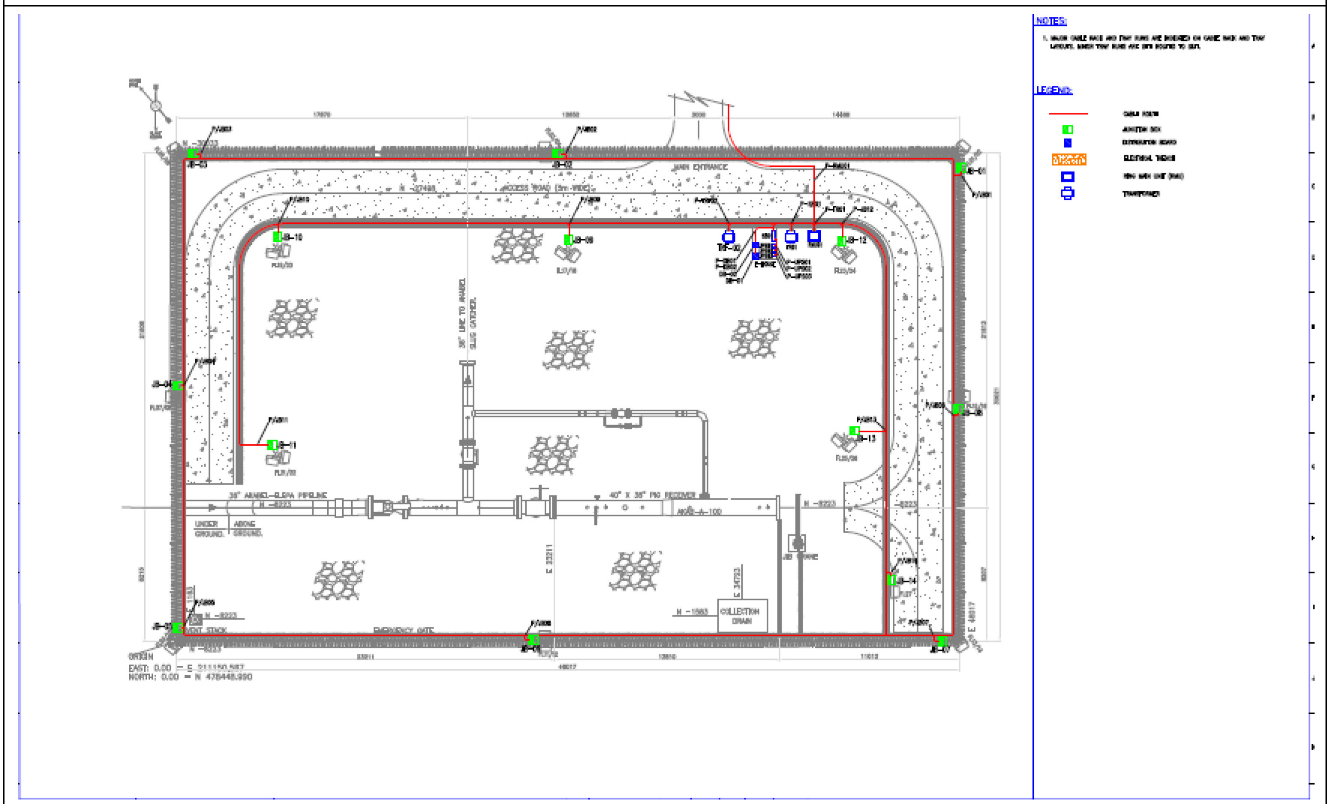
POWER SUPPLY/DISTRIBUTION OPTION 1: ELEPA MANIFOLD STATION



POWER SUPPLY/DISTRIBUTION OPTION 1: AKABEL MANIFOLD STATION



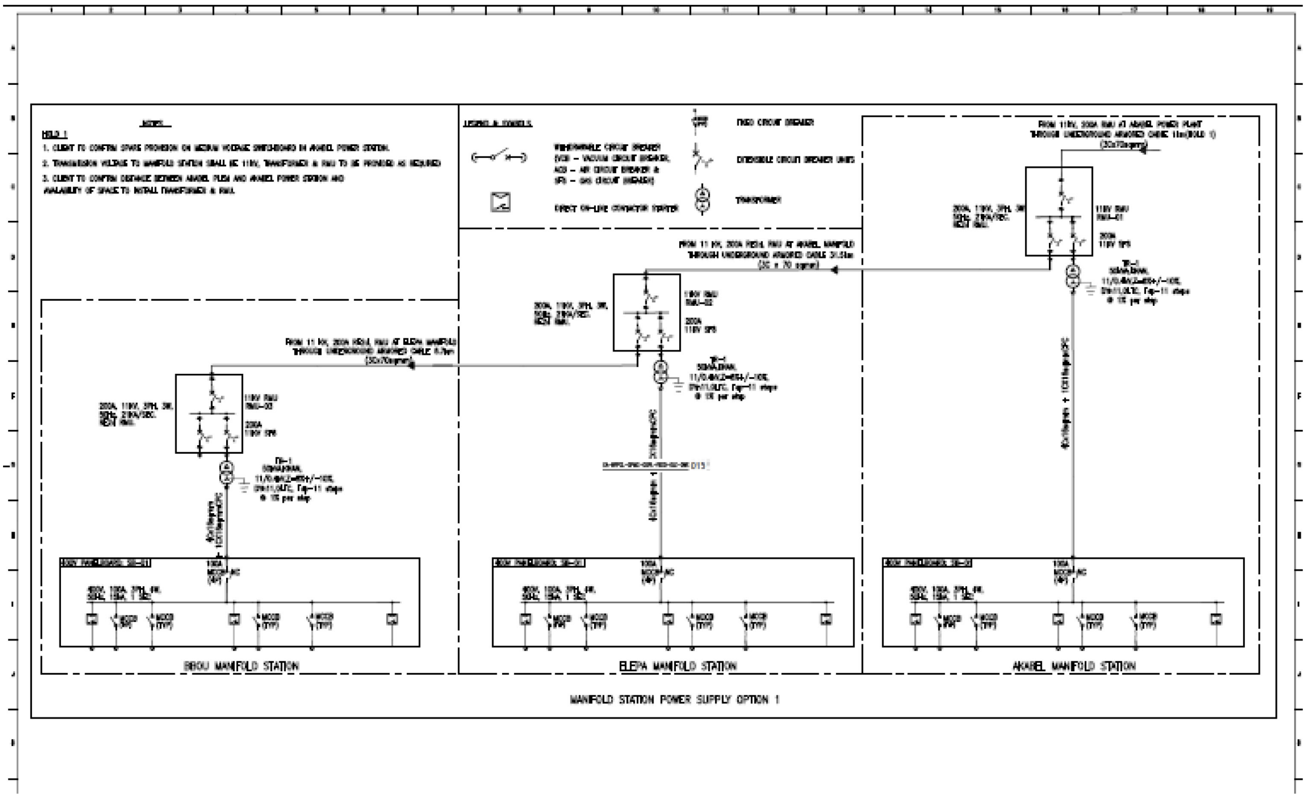
ELEP



AKAB

10.4. Single Line Diagrams

Single Line Diagrams
SLD BBOU, ELEP & AKAB



10.5. Key Electrical Engineering Deliverables

Table 10.1: Key Electrical Engineering Deliverables

S/N	Document No	Document Title	Attachments from Epic	TCE Comments
1	EA-BFPCL-GPMC-GGPL-FEED-ELE-RPT-001	**Finalization of Electrical Power Requirement and Power Supply**	Finalization of Ele Power Req.&Power	EA-BFPCL-GPMC-GGPL-FEED-ELE-RPT-00
2	EA-BFPCL-GPMC-GGPL-	Green Initiatives	Green Initiatives	

	FEED-ELE-RPT-005			
3	EA-BFPCL-GPMC-GGPL-FEED-ELE-DWG-016	Schematics of Power Panels	Sch. of Power Panel (BBOU, ELEP,AKAB)	
4	EA-BFPCL-GPMC-GGPL-FEED-ELE-LST-001	Load & Electrical Equipment List - BBOU, ELEP, AKAB	Load & Ele Equip. List	PDF EA-BFPCL-GPMC-GGPL-FEED-ELE-LST-001
5	EA-BFPCL-GPMC-GGPL-FEED-ELE-LST-005	Material Requisitions for Electrical Long Lead Items	Material Req. for Ele Long Lead Items	
6	EA-BFPCL-GPMC-GGPL-FEED-ELE-LST-006	Distribution Board Schedule	Distribution Board Schedule	
7	EA-BFPCL-GPMC-GGPL-FEED-ELE-DWG-015	Single Line Diagram - Overall, BBOU, ELEP, AKAB	SLD- Power Supply_Distribution	PDF EA-BFPCL-GPMC-GGPL-FEED-ELE-DWG-0
8	EA-BFPCL-GPMC-GGPL-FEED-ELE-DWG-004	Electrical Cable Route Drawings - BBOU, ELEP, AKAB	Cable routing Layout(BBOU,ELE,AK	
9	EA-BFPCL-GPMC-GGPL-FEED-ELE-DWG-007	Lighting Layout - BBOU, ELEP, AKAB	Lighting Layout (BBOU, ELEP,AKAB)	
10	EA-BFPCL-GPMC-GGPL-FEED-ELE-DWG-010	Electrical Power System Layout (Typical)	Typical Pwr Sys Layout(BBOU,ELE,AK	
11	EA-BFPCL-GPMC-GGPL-FEED-ELE-DWG-011	Earthing Layout Drawings - BBOU, ELEP, AKAB	Earthing Layout (BBOU,ELEP,AKAB	
12	EA-BFPCL-GPMC-GGPL-FEED-ELE-DWG-014	Pipeline Cathodic Protection Layout	Ppl cath. protection layout(BBOU,ELE,AK	

13	EA-BFPCL-GPMC-GGPL-FEED-ELE-SPC-002	Specifications for Electrical Equipment	Specification for Electrical Equipment	
14	EA-BFPCL-GPMC-GGPL-FEED-ELE-DSH-002	Data Sheets for Electrical Equipment	Data Sheets for Electrical Equipment	
15	EA-BFPCL-GPMC-GGPL-FEED-ELE-RPT-003	Cable Sizing Calculations	Cable Sizing Calculation Report	
16	EA-BFPCL-GPMC-GGPL-FEED-ELE-RPT-004	Electrical System Sizing Calculation	Electrical System Sizing Calculation	
17	EA-BFPCL-GPMC-GGPL-FEED-ELE-LST-004	Electrical MTOs	Electrical MTOs	
18		Single Line Diagrams	Single Line Diagrams	

11. Control & Instrumentation & Telecommunications Engineering

11.1. Specific Key Drivers / Decissions

The following are specific key drivers / decisions:

1. Design integration between SPDC and BFPCL in respect of Wellhead Manifold Platforms
2. ESD to shut down supply of gas to the pipeline in case of exposure to high pressure
3. Depressurisation to cold vent (for small inventory trap)
4. Data Communication using FOC, with VSAT as back-up.
5. Pipeline leak/intrusion detection using fibre optic cable

11.2. Key Control & Instrumentation and Telecoms Engineering Deliverables

Table 11.1: Key control & Instrumentation and Telecommunications Engineering Deliverables

S/N	Document No.	Document Title	Attachments from EPIC	TCE Comments
1	EA-BFPCL-GPMC-GGPL-FEED-ICS-SPC-023	Instrument Design Basis	Instrument Design Basis	
2	EA-BFPCL-GPMC-GGPL-FEED-ICS-DWG-001	ICSS System Overall System Architecture for BBOU/ELEP/AKAB	ICSS System Overall System Architecture-	
3	EA-BFPCL-GPMC-GGPL-FEED-ICS-DWG-004	ICS System Block Diagrams - BBOU, ELEP, AKAB	Block Diagram-BBOU,ELEP	
4	EA-BFPCL-GPMC-GGPL-FEED-ICS-LST-001	Instrument Point Data Base-BBOU, ELEP, AKAB	Instrument Point DataBase- BBOU,ELE	
5	EA-BFPCL-GPMC-GGPL-FEED-ICS-DSH-001	Specification for Field Guages	Specification for Field Gauges	
6	EA-BFPCL-GPMC-GGPL-FEED-ICS-DSH-002	Specification for Pressure, Temperature and Level Transmitters	Specification for Pressure Temperatu	

S/N	Document No.	Document Title	Attachments from EPIC	TCE Comments
7	EA-BFPCL-GPMC-GGPL-FEED-ICS-DSH-005	Specification for Pipeline Leak Detection System - BBOU/ELEP/AKAB	Specification for Pipeline Leak Detect	
8	EA-BFPCL-GPMC-GGPL-FEED-ICS-DSH-006	Specification for Venturi Based Wet Gas Flow Meters-BBOU/ELEP	Specification for Venturi Based Wet G	
9	EA-BFPCL-GPMC-GGPL-FEED-ICS-DSH-022	Specification for Fire & Gas Detection System-BBOU/ELEP/AKAB	Specification for Fire Gas Detection S	
10	EA-BFPCL-GPMC-GGPL-FEED-ICS-DSH-038	Specification for ESD and BDV Valves-BBOU/ELEP/AKAB	Specification for ESD and BDV Valves_	
11	EA-BFPCL-GPMC-GGPL-FEED-ICS-DSH-039	Specification for Pressure Safety Valves	Specification for Pressure Safety Valv	
12	EA-BFPCL-GPMC-GGPL-FEED-ICS-DSH-040	Specification for Instrument Cable-BBOU/ELEP/AKAB	Specification for Instrument Cable-BE	
13	EA-BFPCL-GPMC-GGPL-FEED-ICS-DSH-041	Specification for Telecommunication-BBOU/ELEP/AKAB	Specification for Telecommunication-	
14	EA-BFPCL-GPMC-GGPL-FEED-ICS-SPC-024	Specifications for Bulk Items-BBOU/ELEP/AKAB	Specifications for Bulk Items for BBOU	
15	EA-BFPCL-GPMC-GGPL-FEED-ICS-LST-009	Cause & Effect Matrix- BBOU, ELEP, AKAB	Cause and Effect Matrix (BBOU,ELEP,A	
16	EA-BFPCL-GPMC-GGPL-FEED-ICS-DWG-019	Typical Loop Diagrams	Typical Loop Diagram	
17	EA-BFPCL-GPMC-GGPL-FEED-ICS-DWG-039	Instrument Hook Up Drawing-BBOU/ELEP/AKAB	Instrument Hook up and Installation	

S/N	Document No.	Document Title	Attachments from EPIC	TCE Comments
18	EA-BFPCL-GPMC-GGPL-FEED-ICS-DWG-025	Instrument Location and Cable Routing Diagram- BBOU, ELEP, AKAB	Instrument Location and Cable	
19	EA-BFPCL-GPMC-GGPL-FEED-ICS-SPC-007	Specification for Junction Box and Cable Gland-BBOU/ELEP/AKAB	Specification for Junction Box and Ca	
20	EA-BFPCL-GPMC-GGPL-FEED-ICS-SPC-012	Specification for SCADA System	Specification for SCADA System	
21	EA-BFPCL-GPMC-GGPL-FEED-ICS-RPT-002	Wake Frequency Calculation	Wake Frequency Calculation	
22	EA-BFPCL-GPMC-GGPL-FEED-ICS-DWG-029	Earthing Philosophy Drawing	Earthing Drawings-BBOU,ELE	
23	EA-BFPCL-GPMC-GGPL-FEED-ICS-LST-007	Instrument MTO's	Instrument MTOs	
24	EA-BFPCL-GPMC-GGPL-FEED-ICS-LST-015	Material Requisitions for Instru-mentation/ Telecommunication Long Lead Items	Material Requisitions for Inst	

12. Technical HSE and Loss Prevention

12.1. Specific Key Drivers / Decissions

The following are specific key drivers / decisions:

1. Design integration between SPDC and BFPCL in respect of Wellhead Manifold Platforms
2. Design that will be environmentally friendly, in respect of containment of inventory, with zero spill to the environment
3. Placement of the facilities at a height that is reasonably above flood level that will ever be experienced in the area, – drawing lessons from the 2012 rare flood experience

12.2. Key Technical Safety and Loss Prevention Deliverables

Table 12.1: Key Technical Safety and Loss Prevention Deliverables

S/N	Description	References	
1	Hazop Report		HAZOP Report
2	Fire and Explosion Philosophy		Fire and Explosion Philosophy (2).pdf
3	HSSE Design Philosophy		HSSE Design Philosophy
4	Hazardous Area Classification Schedule		Hazardous Area Classification Schedul
5	Hazardous Area Classification Drawing		Hazardous Area Classification Drawing
6	Safety Studies & Monsoon Combat Plan		Safety Studies & Monsoon Combat Pla
7	Emergency Response / Disaster Management Plan	Emergency Response Disaster Managemer	
8	Effluent & Emission Requirement & Compliance Plan	Effluent & Emission Requirement & Comp	
9	HAZID Terms of Reference and HAZID Review	HAZID Terms of Reference and HAZID	

S/N	Description	References	
10	Project Hazard & Effect Register (HAZID Report)	Project Hazard & Effect Register (HAZID	
11	HAZOP Terms of Reference and HAZOP Review	HAZOP Terms of Reference and HAZOP	
12	SIL Classification	SIL Classification.pdf	
13	Design ALARP Demonstration (HSE Case)	Design ALARP Demo (HSE Case)	